

Phase I/II Operating Characteristics

Random model - Toxicity prior comparison - allocation = two_stage - IPDE alpha = 0.3 - both utilities - N = 30

	Dose 1	Dose 2	Dose 3	Dose 4	Dose 5	Stop %
Scenario 1						
DLT rate	0.3	0.35	0.4	0.45	0.5	
Efficacy rate	0.05	0.06	0.07	0.08	0.09	
Utility 2	-0.2	-0.3	-0.3	-0.4	-0.4	
Utility 3	31	29.6	28.2	26.8	25.4	
OBD selection (%)						
U2 - Toxicity prior Beta(0.15, 0.85)	34.7	8.4	1	0.3	0.3	55.3
U3 - Toxicity prior Beta(0.15, 0.85)	36.2	7.6	0.3	0.1	0.1	55.7
U2 - Toxicity prior Beta(0.3, 0.7)	33.2	10.3	1.9	0.5	0.6	53.5
U3 - Toxicity prior Beta(0.3, 0.7)	37.3	7.6	0.7	0.2	0.2	54
U2 - Toxicity prior Beta(0.5, 0.5)	33.9	10.4	2	0.7	0.5	52.5
U3 - Toxicity prior Beta(0.5, 0.5)	36.9	9.1	1.2	0.4	0.2	52.2
U2 - Toxicity prior Beta(1, 1)	34	9.2	2.5	0.6	0.5	53.2
U3 - Toxicity prior Beta(1, 1)	38.3	7.4	0.8	0.2	0.1	53.2
Mean patients treated						
U2 - Toxicity prior Beta(0.15, 0.85)	14.8	4.3	2.4	1	0.5	
U3 - Toxicity prior Beta(0.15, 0.85)	14.8	4.2	2.4	1	0.5	
U2 - Toxicity prior Beta(0.3, 0.7)	14.5	4.1	2.7	1.3	0.6	
U3 - Toxicity prior Beta(0.3, 0.7)	14.5	4.2	2.6	1.2	0.6	
U2 - Toxicity prior Beta(0.5, 0.5)	14.3	4	2.8	1.4	0.6	
U3 - Toxicity prior Beta(0.5, 0.5)	14.3	4.1	2.8	1.4	0.6	
U2 - Toxicity prior Beta(1, 1)	14.2	4	2.7	1.5	0.8	
U3 - Toxicity prior Beta(1, 1)	14.2	4	2.7	1.5	0.8	
Mean IPDE patients						
U2 - Toxicity prior Beta(0.15, 0.85)	0.7	2.1	0.6	0.4	0.1	
U3 - Toxicity prior Beta(0.15, 0.85)	0.7	2.1	0.6	0.4	0.1	
U2 - Toxicity prior Beta(0.3, 0.7)	0.7	1.9	0.6	0.5	0.1	
U3 - Toxicity prior Beta(0.3, 0.7)	0.6	2	0.6	0.5	0.1	
U2 - Toxicity prior Beta(0.5, 0.5)	0.6	1.9	0.6	0.4	0.1	
U3 - Toxicity prior Beta(0.5, 0.5)	0.6	1.9	0.6	0.4	0.1	
U2 - Toxicity prior Beta(1, 1)	0.5	1.6	0.5	0.4	0.1	
U3 - Toxicity prior Beta(1, 1)	0.5	1.6	0.5	0.4	0.1	
Scenario 2						
DLT rate	0.15	0.2	0.3	0.35	0.45	
Efficacy rate	0.2	0.25	0.5	0.5	0.5	
Utility 2	0.1	0.1	0.2	0.1	0.1	
Utility 3	46	47	58	56	52	
OBD selection (%)						
U2 - Toxicity prior Beta(0.15, 0.85)	36.4	14.6	32.7	11.4	3.1	1.8
U3 - Toxicity prior Beta(0.15, 0.85)	37.8	16.3	31.3	10.8	2	1.8
U2 - Toxicity prior Beta(0.3, 0.7)	34.7	14.3	32.5	13.2	3.5	1.8
U3 - Toxicity prior Beta(0.3, 0.7)	36.8	15.8	30.9	11.7	3	1.8
U2 - Toxicity prior Beta(0.5, 0.5)	34.2	14.3	32.8	12.9	4	1.8
U3 - Toxicity prior Beta(0.5, 0.5)	36.3	15.8	31.2	11.3	3.6	1.8
U2 - Toxicity prior Beta(1, 1)	32.8	13	34.2	13.9	4.3	1.8
U3 - Toxicity prior Beta(1, 1)	35.8	14.6	30.8	13.9	3.1	1.8
Mean patients treated						
U2 - Toxicity prior Beta(0.15, 0.85)	11.2	5.2	7.1	4.1	2.1	
U3 - Toxicity prior Beta(0.15, 0.85)	11.1	5.3	7.1	4.1	2.1	
U2 - Toxicity prior Beta(0.3, 0.7)	10.7	4.8	6.9	4.8	2.5	
U3 - Toxicity prior Beta(0.3, 0.7)	10.7	4.9	6.9	4.8	2.4	
U2 - Toxicity prior Beta(0.5, 0.5)	10.5	4.7	6.8	5	2.7	
U3 - Toxicity prior Beta(0.5, 0.5)	10.5	4.7	6.8	5	2.7	
U2 - Toxicity prior Beta(1, 1)	10.4	4.7	6.7	5.1	2.9	
U3 - Toxicity prior Beta(1, 1)	10.4	4.7	6.6	5.2	2.9	
Mean IPDE patients						
U2 - Toxicity prior Beta(0.15, 0.85)	0.4	2.3	1.3	0.9	0.3	
U3 - Toxicity prior Beta(0.15, 0.85)	0.4	2.4	1.3	0.9	0.3	
U2 - Toxicity prior Beta(0.3, 0.7)	0.4	2.2	1.3	1	0.4	
U3 - Toxicity prior Beta(0.3, 0.7)	0.4	2.2	1.3	1	0.4	
U2 - Toxicity prior Beta(0.5, 0.5)	0.3	2.2	1.2	1	0.4	
U3 - Toxicity prior Beta(0.5, 0.5)	0.3	2.2	1.2	1	0.3	
U2 - Toxicity prior Beta(1, 1)	0.3	2.1	1.2	1	0.3	
U3 - Toxicity prior Beta(1, 1)	0.3	2.1	1.1	1	0.3	

Phase I/II Operating Characteristics

Random model - Toxicity prior comparison - allocation = two_stage - IPDE alpha = 0.3 - both utilities - N = 30

	Dose 1	Dose 2	Dose 3	Dose 4	Dose 5	Stop %
Scenario 3						
DLT rate	0.15	0.2	0.3	0.35	0.45	
Efficacy rate	0.1	0.4	0.4	0.4	0.4	
Utility 2	-0.1	0.2	0.1	0.1	-0.1	
Utility 3	40	56	52	50	46	
OBD selection (%)						
U2 - Toxicity prior Beta(0.15, 0.85)	18.9	37.2	23.6	8.3	2	10
U3 - Toxicity prior Beta(0.15, 0.85)	20.2	41	20.3	7	1.6	9.9
U2 - Toxicity prior Beta(0.3, 0.7)	18.7	37.1	23.2	9.1	2.3	9.6
U3 - Toxicity prior Beta(0.3, 0.7)	19.9	40.1	21	7.4	2	9.6
U2 - Toxicity prior Beta(0.5, 0.5)	18.8	34.6	24.8	9.1	3.2	9.5
U3 - Toxicity prior Beta(0.5, 0.5)	20.3	37.3	21.8	8.7	2.4	9.5
U2 - Toxicity prior Beta(1, 1)	17.9	33.3	24.9	10.2	4.1	9.6
U3 - Toxicity prior Beta(1, 1)	20.3	36.1	22.1	9	3	9.5
Mean patients treated						
U2 - Toxicity prior Beta(0.15, 0.85)	9.3	7.3	6.3	3.9	2.1	
U3 - Toxicity prior Beta(0.15, 0.85)	9.3	7.3	6.3	3.9	2.1	
U2 - Toxicity prior Beta(0.3, 0.7)	9.1	6.7	6.2	4.5	2.5	
U3 - Toxicity prior Beta(0.3, 0.7)	9.1	6.8	6.2	4.5	2.4	
U2 - Toxicity prior Beta(0.5, 0.5)	8.9	6.5	6.2	4.8	2.7	
U3 - Toxicity prior Beta(0.5, 0.5)	9	6.5	6.2	4.8	2.6	
U2 - Toxicity prior Beta(1, 1)	8.9	6.5	6	4.7	2.9	
U3 - Toxicity prior Beta(1, 1)	8.9	6.6	6	4.7	2.9	
Mean IPDE patients						
U2 - Toxicity prior Beta(0.15, 0.85)	0.2	2.8	1.3	1.1	0.4	
U3 - Toxicity prior Beta(0.15, 0.85)	0.2	2.8	1.2	1.1	0.4	
U2 - Toxicity prior Beta(0.3, 0.7)	0.2	2.7	1.3	1.2	0.4	
U3 - Toxicity prior Beta(0.3, 0.7)	0.2	2.7	1.2	1.2	0.4	
U2 - Toxicity prior Beta(0.5, 0.5)	0.2	2.6	1.2	1.1	0.4	
U3 - Toxicity prior Beta(0.5, 0.5)	0.2	2.6	1.2	1.1	0.4	
U2 - Toxicity prior Beta(1, 1)	0.2	2.6	1.1	1.1	0.3	
U3 - Toxicity prior Beta(1, 1)	0.2	2.6	1.1	1.1	0.3	
Scenario 4						
DLT rate	0.05	0.1	0.18	0.3	0.4	
Efficacy rate	0.05	0.1	0.25	0.5	0.55	
Utility 2	0	0	0.1	0.2	0.1	
Utility 3	41	42	47.8	58	57	
OBD selection (%)						
U2 - Toxicity prior Beta(0.15, 0.85)	9.3	7.5	22.5	39.4	19.4	1.9
U3 - Toxicity prior Beta(0.15, 0.85)	15.2	8.6	20.4	38.3	15.7	1.8
U2 - Toxicity prior Beta(0.3, 0.7)	8	6.5	20.3	43.4	20	1.8
U3 - Toxicity prior Beta(0.3, 0.7)	13	8.4	18.9	40.9	17	1.8
U2 - Toxicity prior Beta(0.5, 0.5)	7.6	5.8	19.3	43.9	21.5	1.9
U3 - Toxicity prior Beta(0.5, 0.5)	11.3	7.8	17.6	42.8	18.6	1.9
U2 - Toxicity prior Beta(1, 1)	7.2	5.8	18.2	43.1	23.8	1.9
U3 - Toxicity prior Beta(1, 1)	11.2	7.7	16.5	42.8	20	1.8
Mean patients treated						
U2 - Toxicity prior Beta(0.15, 0.85)	4.4	4.1	6.5	8	6.8	
U3 - Toxicity prior Beta(0.15, 0.85)	4.4	4.2	6.5	7.9	6.9	
U2 - Toxicity prior Beta(0.3, 0.7)	4.3	3.9	6.1	8.3	7.2	
U3 - Toxicity prior Beta(0.3, 0.7)	4.2	3.9	6	8.4	7.2	
U2 - Toxicity prior Beta(0.5, 0.5)	4.2	3.8	5.8	8.5	7.4	
U3 - Toxicity prior Beta(0.5, 0.5)	4.2	3.8	5.8	8.5	7.5	
U2 - Toxicity prior Beta(1, 1)	4.2	3.7	5.7	8.2	7.9	
U3 - Toxicity prior Beta(1, 1)	4.2	3.7	5.6	8.4	7.9	
Mean IPDE patients						
U2 - Toxicity prior Beta(0.15, 0.85)	0.3	2.8	1.4	2.3	0.8	
U3 - Toxicity prior Beta(0.15, 0.85)	0.3	2.8	1.4	2.3	0.8	
U2 - Toxicity prior Beta(0.3, 0.7)	0.2	2.7	1.4	2.3	0.8	
U3 - Toxicity prior Beta(0.3, 0.7)	0.2	2.7	1.4	2.2	0.8	
U2 - Toxicity prior Beta(0.5, 0.5)	0.2	2.6	1.4	2.2	0.8	
U3 - Toxicity prior Beta(0.5, 0.5)	0.2	2.6	1.4	2.2	0.8	
U2 - Toxicity prior Beta(1, 1)	0.2	2.6	1.3	2.2	0.7	
U3 - Toxicity prior Beta(1, 1)	0.2	2.6	1.3	2.2	0.7	

Phase I/II Operating Characteristics

Random model - Toxicity prior comparison - allocation = two_stage - IPDE alpha = 0.3 - both utilities - N = 30

	Dose 1	Dose 2	Dose 3	Dose 4	Dose 5	Stop %
Scenario 5						
DLT rate	0.07	0.12	0.17	0.22	0.3	
Efficacy rate	0.05	0.3	0.3	0.3	0.3	
Utility 2	0	0.2	0.1	0.1	0	
Utility 3	40.2	53.2	51.2	49.2	46	
OBD selection (%)						
U2 - Toxicity prior Beta(0.15, 0.85)	5.1	29.5	29.4	20	11.8	4.2
U3 - Toxicity prior Beta(0.15, 0.85)	9.6	35.3	26.2	17.7	7	4.2
U2 - Toxicity prior Beta(0.3, 0.7)	5.1	28.1	28.8	21.9	11.9	4.2
U3 - Toxicity prior Beta(0.3, 0.7)	9.5	34.6	25.5	19.3	6.9	4.2
U2 - Toxicity prior Beta(0.5, 0.5)	5	28.9	26.2	22.7	13	4.2
U3 - Toxicity prior Beta(0.5, 0.5)	8.9	35.2	24.9	19.1	7.7	4.2
U2 - Toxicity prior Beta(1, 1)	4.7	27.8	27.1	22.5	13.7	4.2
U3 - Toxicity prior Beta(1, 1)	8.6	34.4	24.8	19.7	8.3	4.2
Mean patients treated						
U2 - Toxicity prior Beta(0.15, 0.85)	4.8	6.1	6.6	5.8	6.1	
U3 - Toxicity prior Beta(0.15, 0.85)	4.9	5.9	6.7	5.8	6.2	
U2 - Toxicity prior Beta(0.3, 0.7)	4.8	5.8	6.5	6	6.4	
U3 - Toxicity prior Beta(0.3, 0.7)	4.8	5.7	6.6	6	6.4	
U2 - Toxicity prior Beta(0.5, 0.5)	4.8	5.7	6.3	6.2	6.6	
U3 - Toxicity prior Beta(0.5, 0.5)	4.8	5.7	6.3	6.2	6.5	
U2 - Toxicity prior Beta(1, 1)	4.7	5.6	6.2	6.1	6.8	
U3 - Toxicity prior Beta(1, 1)	4.7	5.6	6.2	6.1	6.8	
Mean IPDE patients						
U2 - Toxicity prior Beta(0.15, 0.85)	0.2	3.2	1.7	2.2	1.1	
U3 - Toxicity prior Beta(0.15, 0.85)	0.2	3.2	1.7	2.2	1.1	
U2 - Toxicity prior Beta(0.3, 0.7)	0.2	3.2	1.7	2.2	1.1	
U3 - Toxicity prior Beta(0.3, 0.7)	0.2	3.2	1.7	2.2	1.1	
U2 - Toxicity prior Beta(0.5, 0.5)	0.2	3.1	1.7	2.2	1.1	
U3 - Toxicity prior Beta(0.5, 0.5)	0.2	3.1	1.7	2.2	1.1	
U2 - Toxicity prior Beta(1, 1)	0.2	3.1	1.7	2.2	1.1	
U3 - Toxicity prior Beta(1, 1)	0.2	3.1	1.7	2.1	1.1	
Scenario 6						
DLT rate	0.29	0.34	0.38	0.43	0.52	
Efficacy rate	0.3	0.32	0.35	0.38	0.4	
Utility 2	0	0	0	-0.1	-0.1	
Utility 3	46.4	45.6	45.8	45.6	43.2	
OBD selection (%)						
U2 - Toxicity prior Beta(0.15, 0.85)	69	12.4	3.5	1	0.4	13.7
U3 - Toxicity prior Beta(0.15, 0.85)	69.1	12.6	2.8	1.3	0.5	13.7
U2 - Toxicity prior Beta(0.3, 0.7)	67.9	12.7	4.3	1.4	0.8	12.9
U3 - Toxicity prior Beta(0.3, 0.7)	70.1	11.9	3.4	0.9	0.4	13.3
U2 - Toxicity prior Beta(0.5, 0.5)	65.8	13.2	4.9	1.8	1.1	13.2
U3 - Toxicity prior Beta(0.5, 0.5)	69.7	11.8	3.7	1.6	0.3	12.9
U2 - Toxicity prior Beta(1, 1)	66.2	13.4	4.8	1.7	1	12.9
U3 - Toxicity prior Beta(1, 1)	68	13.2	3.9	1.4	0.5	13
Mean patients treated						
U2 - Toxicity prior Beta(0.15, 0.85)	19.1	4.3	2.4	1.2	0.6	
U3 - Toxicity prior Beta(0.15, 0.85)	18.9	4.4	2.5	1.2	0.6	
U2 - Toxicity prior Beta(0.3, 0.7)	18.6	4.1	2.8	1.4	0.7	
U3 - Toxicity prior Beta(0.3, 0.7)	18.5	4.1	2.8	1.4	0.7	
U2 - Toxicity prior Beta(0.5, 0.5)	18.3	4.1	2.8	1.6	0.8	
U3 - Toxicity prior Beta(0.5, 0.5)	18.3	4.2	2.8	1.6	0.8	
U2 - Toxicity prior Beta(1, 1)	18.2	4.2	2.8	1.6	0.8	
U3 - Toxicity prior Beta(1, 1)	18.2	4.2	2.7	1.6	0.8	
Mean IPDE patients						
U2 - Toxicity prior Beta(0.15, 0.85)	0.4	1.5	0.5	0.3	0.1	
U3 - Toxicity prior Beta(0.15, 0.85)	0.4	1.6	0.5	0.3	0.1	
U2 - Toxicity prior Beta(0.3, 0.7)	0.5	1.5	0.5	0.4	0.1	
U3 - Toxicity prior Beta(0.3, 0.7)	0.5	1.5	0.5	0.4	0.1	
U2 - Toxicity prior Beta(0.5, 0.5)	0.4	1.4	0.5	0.3	0.1	
U3 - Toxicity prior Beta(0.5, 0.5)	0.4	1.4	0.5	0.3	0.1	
U2 - Toxicity prior Beta(1, 1)	0.3	1.3	0.4	0.3	0.1	
U3 - Toxicity prior Beta(1, 1)	0.3	1.3	0.4	0.3	0.1	

Phase I/II Operating Characteristics

Random model - Toxicity prior comparison - allocation = two_stage - IPDE alpha = 0.3 - both utilities - N = 30

	Dose 1	Dose 2	Dose 3	Dose 4	Dose 5	Stop %
Scenario 7						
DLT rate	0.16	0.2	0.26	0.31	0.36	
Efficacy rate	0.2	0.5	0.5	0.5	0.5	
Utility 2	0	0.3	0.2	0.2	0.1	
Utility 3	45.6	62	59.6	57.6	55.6	
OBD selection (%)						
U2 - Toxicity prior Beta(0.15, 0.85)	26.7	34.2	21.4	10.9	4.4	2.4
U3 - Toxicity prior Beta(0.15, 0.85)	27.6	37.6	21.3	8.9	2.3	2.3
U2 - Toxicity prior Beta(0.3, 0.7)	26.3	31.7	23	11.7	5.1	2.2
U3 - Toxicity prior Beta(0.3, 0.7)	28.2	34.9	21.2	9.7	3.8	2.2
U2 - Toxicity prior Beta(0.5, 0.5)	26.2	30	23.3	12.8	5.5	2.2
U3 - Toxicity prior Beta(0.5, 0.5)	27.4	35.1	21.1	9.9	4.3	2.2
U2 - Toxicity prior Beta(1, 1)	25.9	30.9	22.8	12.6	5.6	2.2
U3 - Toxicity prior Beta(1, 1)	27.4	35.6	20.6	10.2	4	2.2
Mean patients treated						
U2 - Toxicity prior Beta(0.15, 0.85)	10.5	6.7	5.8	3.9	2.7	
U3 - Toxicity prior Beta(0.15, 0.85)	10.4	6.8	6	3.9	2.6	
U2 - Toxicity prior Beta(0.3, 0.7)	10.1	6.3	5.8	4.4	3.1	
U3 - Toxicity prior Beta(0.3, 0.7)	10.1	6.3	5.7	4.5	3.1	
U2 - Toxicity prior Beta(0.5, 0.5)	10	6.1	5.8	4.5	3.3	
U3 - Toxicity prior Beta(0.5, 0.5)	10	6.2	5.7	4.4	3.3	
U2 - Toxicity prior Beta(1, 1)	10	6.2	5.6	4.4	3.4	
U3 - Toxicity prior Beta(1, 1)	10	6.2	5.6	4.5	3.4	
Mean IPDE patients						
U2 - Toxicity prior Beta(0.15, 0.85)	0.2	2.5	1.1	1	0.4	
U3 - Toxicity prior Beta(0.15, 0.85)	0.2	2.5	1.1	1	0.4	
U2 - Toxicity prior Beta(0.3, 0.7)	0.2	2.5	1.1	1.1	0.4	
U3 - Toxicity prior Beta(0.3, 0.7)	0.2	2.4	1.1	1.1	0.4	
U2 - Toxicity prior Beta(0.5, 0.5)	0.2	2.4	1.1	1	0.4	
U3 - Toxicity prior Beta(0.5, 0.5)	0.2	2.4	1.1	1	0.4	
U2 - Toxicity prior Beta(1, 1)	0.2	2.4	1	1	0.4	
U3 - Toxicity prior Beta(1, 1)	0.2	2.4	1.1	1	0.4	
Scenario 8						
DLT rate	0.32	0.37	0.42	0.48	0.52	
Efficacy rate	0.1	0.15	0.2	0.3	0.4	
Utility 2	-0.2	-0.2	-0.2	-0.2	-0.1	
Utility 3	33.2	34.2	35.2	38.8	43.2	
OBD selection (%)						
U2 - Toxicity prior Beta(0.15, 0.85)	44.3	8.1	2.5	1.5	0.4	43.2
U3 - Toxicity prior Beta(0.15, 0.85)	46.7	7.2	1.7	1.1	0.4	42.9
U2 - Toxicity prior Beta(0.3, 0.7)	43.5	9.1	3.2	1.7	0.6	41.9
U3 - Toxicity prior Beta(0.3, 0.7)	45.2	8.6	2.7	1.3	0.3	41.9
U2 - Toxicity prior Beta(0.5, 0.5)	43.2	9.1	3.7	1.7	0.6	41.7
U3 - Toxicity prior Beta(0.5, 0.5)	45.8	8.3	2.4	1.5	0.4	41.6
U2 - Toxicity prior Beta(1, 1)	43	8.7	3.7	2	1.1	41.5
U3 - Toxicity prior Beta(1, 1)	45.1	8	2.6	1.8	0.8	41.7
Mean patients treated						
U2 - Toxicity prior Beta(0.15, 0.85)	16.8	4	2.1	0.8	0.4	
U3 - Toxicity prior Beta(0.15, 0.85)	16.8	4	2.2	0.8	0.4	
U2 - Toxicity prior Beta(0.3, 0.7)	16.5	3.8	2.5	1.1	0.5	
U3 - Toxicity prior Beta(0.3, 0.7)	16.5	3.8	2.4	1.1	0.5	
U2 - Toxicity prior Beta(0.5, 0.5)	16.4	3.7	2.5	1.3	0.6	
U3 - Toxicity prior Beta(0.5, 0.5)	16.4	3.7	2.5	1.3	0.5	
U2 - Toxicity prior Beta(1, 1)	16.3	3.7	2.4	1.3	0.6	
U3 - Toxicity prior Beta(1, 1)	16.3	3.7	2.4	1.3	0.6	
Mean IPDE patients						
U2 - Toxicity prior Beta(0.15, 0.85)	0.4	1.6	0.4	0.2	0.1	
U3 - Toxicity prior Beta(0.15, 0.85)	0.4	1.6	0.4	0.2	0.1	
U2 - Toxicity prior Beta(0.3, 0.7)	0.4	1.6	0.5	0.3	0.1	
U3 - Toxicity prior Beta(0.3, 0.7)	0.4	1.6	0.5	0.3	0.1	
U2 - Toxicity prior Beta(0.5, 0.5)	0.3	1.5	0.4	0.3	0	
U3 - Toxicity prior Beta(0.5, 0.5)	0.3	1.5	0.4	0.3	0	
U2 - Toxicity prior Beta(1, 1)	0.3	1.3	0.4	0.2	0	
U3 - Toxicity prior Beta(1, 1)	0.3	1.3	0.4	0.3	0	

Phase I/II Operating Characteristics

Random model - Toxicity prior comparison - allocation = two_stage - IPDE alpha = 0.3 - both utilities - N = 30

	Dose 1	Dose 2	Dose 3	Dose 4	Dose 5	Stop %
Scenario 9						
DLT rate	0.15	0.18	0.22	0.26	0.29	
Efficacy rate	0.05	0.08	0.2	0.25	0.5	
Utility 2	-0.1	-0.1	0	0	0.2	
Utility 3	37	37.6	43.2	44.6	58.4	
OBD selection (%)						
U2 - Toxicity prior Beta(0.15, 0.85)	17.1	10.6	15.9	11	27	18.4
U3 - Toxicity prior Beta(0.15, 0.85)	21.7	11.9	14.4	10.3	23.3	18.4
U2 - Toxicity prior Beta(0.3, 0.7)	14.8	8.5	15.6	12.9	30	18.2
U3 - Toxicity prior Beta(0.3, 0.7)	19.6	10.8	14.3	11.6	25.7	18
U2 - Toxicity prior Beta(0.5, 0.5)	13.5	8.1	14.4	13.9	32.2	17.9
U3 - Toxicity prior Beta(0.5, 0.5)	17.6	10.8	13.3	12.4	27.9	18
U2 - Toxicity prior Beta(1, 1)	12.8	6.5	13.4	14.5	34.9	17.9
U3 - Toxicity prior Beta(1, 1)	18	8.7	11.3	13.8	30.3	17.9
Mean patients treated						
U2 - Toxicity prior Beta(0.15, 0.85)	8.7	4.4	5.4	3.8	5.6	
U3 - Toxicity prior Beta(0.15, 0.85)	8.7	4.4	5.3	3.9	5.6	
U2 - Toxicity prior Beta(0.3, 0.7)	8.4	3.9	5.1	4.4	6.2	
U3 - Toxicity prior Beta(0.3, 0.7)	8.4	3.9	5.1	4.4	6.1	
U2 - Toxicity prior Beta(0.5, 0.5)	8.2	3.7	5	4.6	6.6	
U3 - Toxicity prior Beta(0.5, 0.5)	8.2	3.7	5	4.5	6.5	
U2 - Toxicity prior Beta(1, 1)	8	3.7	4.7	4.5	7.1	
U3 - Toxicity prior Beta(1, 1)	8.1	3.6	4.6	4.5	7.1	
Mean IPDE patients						
U2 - Toxicity prior Beta(0.15, 0.85)	0.5	2.6	1.3	1.4	0.7	
U3 - Toxicity prior Beta(0.15, 0.85)	0.5	2.6	1.3	1.4	0.7	
U2 - Toxicity prior Beta(0.3, 0.7)	0.4	2.4	1.4	1.5	0.7	
U3 - Toxicity prior Beta(0.3, 0.7)	0.4	2.4	1.4	1.5	0.7	
U2 - Toxicity prior Beta(0.5, 0.5)	0.3	2.4	1.3	1.4	0.7	
U3 - Toxicity prior Beta(0.5, 0.5)	0.3	2.4	1.3	1.4	0.7	
U2 - Toxicity prior Beta(1, 1)	0.3	2.3	1.2	1.4	0.7	
U3 - Toxicity prior Beta(1, 1)	0.3	2.3	1.2	1.5	0.7	
Scenario 10						
DLT rate	0.23	0.28	0.32	0.39	0.43	
Efficacy rate	0.05	0.4	0.42	0.45	0.5	
Utility 2	-0.2	0.1	0.1	0.1	0.1	
Utility 3	33.8	52.8	52.4	51.4	52.8	
OBD selection (%)						
U2 - Toxicity prior Beta(0.15, 0.85)	13.2	28.9	12.2	6.1	2.7	36.9
U3 - Toxicity prior Beta(0.15, 0.85)	13.8	30.2	11.2	5.7	2.2	36.9
U2 - Toxicity prior Beta(0.3, 0.7)	11.6	27.4	15.5	7	2.3	36.2
U3 - Toxicity prior Beta(0.3, 0.7)	12.8	29.8	13.8	5.6	2	36
U2 - Toxicity prior Beta(0.5, 0.5)	11.4	27.8	15.2	6.6	3.3	35.7
U3 - Toxicity prior Beta(0.5, 0.5)	11.6	29.4	13.7	6.7	2.4	36.2
U2 - Toxicity prior Beta(1, 1)	11.2	26.1	16	7.3	3.6	35.8
U3 - Toxicity prior Beta(1, 1)	11.4	29.4	13.4	7.1	3	35.7
Mean patients treated						
U2 - Toxicity prior Beta(0.15, 0.85)	11.2	6.3	4.1	2.4	1.5	
U3 - Toxicity prior Beta(0.15, 0.85)	11.2	6.2	4.2	2.4	1.5	
U2 - Toxicity prior Beta(0.3, 0.7)	11	5.6	4.5	2.7	1.7	
U3 - Toxicity prior Beta(0.3, 0.7)	11	5.8	4.5	2.7	1.7	
U2 - Toxicity prior Beta(0.5, 0.5)	10.6	5.5	4.6	3	1.9	
U3 - Toxicity prior Beta(0.5, 0.5)	10.7	5.5	4.6	3	1.9	
U2 - Toxicity prior Beta(1, 1)	10.6	5.6	4.5	3	2.1	
U3 - Toxicity prior Beta(1, 1)	10.6	5.6	4.4	2.9	2.1	
Mean IPDE patients						
U2 - Toxicity prior Beta(0.15, 0.85)	0.1	2.3	0.8	0.6	0.2	
U3 - Toxicity prior Beta(0.15, 0.85)	0.1	2.3	0.8	0.6	0.2	
U2 - Toxicity prior Beta(0.3, 0.7)	0.1	2.1	0.9	0.7	0.2	
U3 - Toxicity prior Beta(0.3, 0.7)	0.1	2.1	0.9	0.7	0.2	
U2 - Toxicity prior Beta(0.5, 0.5)	0.1	2	0.8	0.7	0.2	
U3 - Toxicity prior Beta(0.5, 0.5)	0.1	2	0.8	0.6	0.2	
U2 - Toxicity prior Beta(1, 1)	0.1	1.9	0.8	0.6	0.2	
U3 - Toxicity prior Beta(1, 1)	0.1	1.9	0.8	0.6	0.2	

Phase I/II Operating Characteristics

Random model - Toxicity prior comparison - allocation = two_stage - IPDE alpha = 0.3 - both utilities - N = 30

	Dose 1	Dose 2	Dose 3	Dose 4	Dose 5	Stop %
Scenario 11						
DLT rate	0.22	0.32	0.44	0.59	0.67	
Efficacy rate	0.2	0.2	0.2	0.2	0.2	
Utility 2	0	-0.1	-0.2	-0.4	-0.5	
Utility 3	43.2	39.2	34.4	28.4	25.2	
OBD selection (%)						
U2 - Toxicity prior Beta(0.15, 0.85)	68.1	20.4	3.7	0.6	0	7.2
U3 - Toxicity prior Beta(0.15, 0.85)	72.4	17.7	2.5	0.4	0	7
U2 - Toxicity prior Beta(0.3, 0.7)	69.8	18.5	4.2	0.7	0	6.8
U3 - Toxicity prior Beta(0.3, 0.7)	72.4	17.6	2.7	0.5	0	6.8
U2 - Toxicity prior Beta(0.5, 0.5)	67.4	19.7	5.5	0.6	0.1	6.7
U3 - Toxicity prior Beta(0.5, 0.5)	71.5	17.2	3.9	0.5	0.1	6.8
U2 - Toxicity prior Beta(1, 1)	68.8	18.8	4.6	1	0.1	6.7
U3 - Toxicity prior Beta(1, 1)	72.2	16.6	3.7	0.9	0.1	6.5
Mean patients treated						
U2 - Toxicity prior Beta(0.15, 0.85)	17.8	6.2	3.5	1.2	0.3	
U3 - Toxicity prior Beta(0.15, 0.85)	17.8	6.2	3.5	1.2	0.3	
U2 - Toxicity prior Beta(0.3, 0.7)	17.4	6	3.7	1.5	0.3	
U3 - Toxicity prior Beta(0.3, 0.7)	17.4	6	3.8	1.5	0.3	
U2 - Toxicity prior Beta(0.5, 0.5)	17	6.1	3.8	1.7	0.3	
U3 - Toxicity prior Beta(0.5, 0.5)	17.1	5.9	3.9	1.7	0.3	
U2 - Toxicity prior Beta(1, 1)	17.2	5.9	3.8	1.7	0.4	
U3 - Toxicity prior Beta(1, 1)	17.1	5.8	3.9	1.8	0.4	
Mean IPDE patients						
U2 - Toxicity prior Beta(0.15, 0.85)	0.7	2.4	0.7	0.4	0	
U3 - Toxicity prior Beta(0.15, 0.85)	0.7	2.4	0.7	0.4	0	
U2 - Toxicity prior Beta(0.3, 0.7)	0.8	2.3	0.8	0.5	0	
U3 - Toxicity prior Beta(0.3, 0.7)	0.8	2.3	0.8	0.5	0	
U2 - Toxicity prior Beta(0.5, 0.5)	0.7	2.2	0.7	0.4	0	
U3 - Toxicity prior Beta(0.5, 0.5)	0.7	2.2	0.7	0.4	0	
U2 - Toxicity prior Beta(1, 1)	0.6	2	0.6	0.4	0	
U3 - Toxicity prior Beta(1, 1)	0.6	2	0.6	0.4	0	
Scenario 12						
DLT rate	0.06	0.09	0.18	0.29	0.44	
Efficacy rate	0.05	0.1	0.12	0.4	0.6	
Utility 2	0	0	-0.1	0.1	0.2	
Utility 3	40.6	42.4	40	52.4	58.4	
OBD selection (%)						
U2 - Toxicity prior Beta(0.15, 0.85)	18.1	13.2	10.5	32.2	22.7	3.3
U3 - Toxicity prior Beta(0.15, 0.85)	24.2	13.4	8.1	31.2	19.9	3.2
U2 - Toxicity prior Beta(0.3, 0.7)	15.6	11.4	10.2	35.6	24	3.2
U3 - Toxicity prior Beta(0.3, 0.7)	23.1	11.3	8	32.3	22.1	3.2
U2 - Toxicity prior Beta(0.5, 0.5)	14.4	10.8	9.9	36.7	25	3.2
U3 - Toxicity prior Beta(0.5, 0.5)	20.3	11.5	7.3	35.3	22.3	3.3
U2 - Toxicity prior Beta(1, 1)	14.7	10.1	10.3	34.7	27	3.2
U3 - Toxicity prior Beta(1, 1)	20.7	11.9	7.2	33.4	23.5	3.3
Mean patients treated						
U2 - Toxicity prior Beta(0.15, 0.85)	5	4.6	5.2	7.2	7.6	
U3 - Toxicity prior Beta(0.15, 0.85)	5.1	4.6	5.2	7.2	7.5	
U2 - Toxicity prior Beta(0.3, 0.7)	4.9	4.1	5.1	7.5	8	
U3 - Toxicity prior Beta(0.3, 0.7)	4.8	4.2	5.1	7.5	8	
U2 - Toxicity prior Beta(0.5, 0.5)	4.7	4	4.9	7.6	8.3	
U3 - Toxicity prior Beta(0.5, 0.5)	4.7	4	4.8	7.7	8.3	
U2 - Toxicity prior Beta(1, 1)	4.7	3.9	4.6	7.5	8.9	
U3 - Toxicity prior Beta(1, 1)	4.7	4	4.6	7.4	8.8	
Mean IPDE patients						
U2 - Toxicity prior Beta(0.15, 0.85)	0.4	2.9	1.4	2.2	0.8	
U3 - Toxicity prior Beta(0.15, 0.85)	0.4	2.9	1.4	2.2	0.8	
U2 - Toxicity prior Beta(0.3, 0.7)	0.3	2.7	1.4	2.2	0.8	
U3 - Toxicity prior Beta(0.3, 0.7)	0.3	2.7	1.4	2.2	0.8	
U2 - Toxicity prior Beta(0.5, 0.5)	0.3	2.7	1.3	2.2	0.9	
U3 - Toxicity prior Beta(0.5, 0.5)	0.3	2.7	1.3	2.2	0.8	
U2 - Toxicity prior Beta(1, 1)	0.3	2.7	1.3	2.1	0.8	
U3 - Toxicity prior Beta(1, 1)	0.3	2.7	1.3	2.1	0.8	

Phase I/II Operating Characteristics

Random model - Toxicity prior comparison - allocation = two_stage - IPDE alpha = 0.3 - both utilities - N = 30

	Dose 1	Dose 2	Dose 3	Dose 4	Dose 5	Stop %
Scenario 13						
DLT rate	0.21	0.29	0.3	0.39	0.46	
Efficacy rate	0.05	0.3	0.32	0.35	0.4	
Utility 2	-0.2	0	0	0	-0.1	
Utility 3	34.6	46.4	47.2	45.4	45.6	
OBD selection (%)						
U2 - Toxicity prior Beta(0.15, 0.85)	16.5	29.8	13.5	5.9	2.5	31.8
U3 - Toxicity prior Beta(0.15, 0.85)	17.8	31.7	11.6	5.2	1.7	32
U2 - Toxicity prior Beta(0.3, 0.7)	15.5	29.1	14.6	7.3	2.6	30.9
U3 - Toxicity prior Beta(0.3, 0.7)	17.8	30	12.4	6.4	2.3	31.1
U2 - Toxicity prior Beta(0.5, 0.5)	15	28.1	16.2	6.8	3	30.9
U3 - Toxicity prior Beta(0.5, 0.5)	16.9	29.6	14.1	5.7	2.8	30.9
U2 - Toxicity prior Beta(1, 1)	14.7	27.7	15.6	7.2	3.6	31.2
U3 - Toxicity prior Beta(1, 1)	16.9	28.8	13.9	6.2	3	31.2
Mean patients treated						
U2 - Toxicity prior Beta(0.15, 0.85)	10.9	6.3	4.5	2.7	1.6	
U3 - Toxicity prior Beta(0.15, 0.85)	10.9	6.3	4.5	2.7	1.6	
U2 - Toxicity prior Beta(0.3, 0.7)	10.5	6	4.9	3.1	1.8	
U3 - Toxicity prior Beta(0.3, 0.7)	10.5	6	4.8	3.1	1.9	
U2 - Toxicity prior Beta(0.5, 0.5)	10.3	5.7	4.9	3.3	2.1	
U3 - Toxicity prior Beta(0.5, 0.5)	10.3	5.7	4.9	3.3	2.1	
U2 - Toxicity prior Beta(1, 1)	10.2	5.7	4.7	3.4	2.3	
U3 - Toxicity prior Beta(1, 1)	10.2	5.8	4.8	3.4	2.2	
Mean IPDE patients						
U2 - Toxicity prior Beta(0.15, 0.85)	0.2	2.6	0.9	0.8	0.3	
U3 - Toxicity prior Beta(0.15, 0.85)	0.2	2.6	0.9	0.8	0.3	
U2 - Toxicity prior Beta(0.3, 0.7)	0.2	2.4	1	0.8	0.3	
U3 - Toxicity prior Beta(0.3, 0.7)	0.2	2.4	1	0.8	0.3	
U2 - Toxicity prior Beta(0.5, 0.5)	0.2	2.3	0.9	0.8	0.2	
U3 - Toxicity prior Beta(0.5, 0.5)	0.2	2.3	1	0.8	0.2	
U2 - Toxicity prior Beta(1, 1)	0.2	2.2	0.9	0.8	0.2	
U3 - Toxicity prior Beta(1, 1)	0.1	2.2	0.9	0.8	0.2	
Scenario 14						
DLT rate	0.09	0.15	0.23	0.28	0.39	
Efficacy rate	0.05	0.08	0.12	0.3	0.4	
Utility 2	0	-0.1	-0.1	0	0	
Utility 3	39.4	38.8	38	46.8	48.4	
OBD selection (%)						
U2 - Toxicity prior Beta(0.15, 0.85)	22.8	15.5	13.5	24.6	16.4	7.2
U3 - Toxicity prior Beta(0.15, 0.85)	32.6	16.6	10.2	21.3	12	7.3
U2 - Toxicity prior Beta(0.3, 0.7)	21.5	13.9	11.8	27	18.7	7.1
U3 - Toxicity prior Beta(0.3, 0.7)	29.9	15.8	9.3	24.1	13.8	7.1
U2 - Toxicity prior Beta(0.5, 0.5)	19.6	12.8	12.1	28.7	19.7	7.1
U3 - Toxicity prior Beta(0.5, 0.5)	28.8	14	9.6	26.6	14	7
U2 - Toxicity prior Beta(1, 1)	18.8	11.6	11.4	30.2	20.9	7.1
U3 - Toxicity prior Beta(1, 1)	28	12.9	9.8	26.7	15.5	7.1
Mean patients treated						
U2 - Toxicity prior Beta(0.15, 0.85)	6.8	5.3	5.6	5.9	5.6	
U3 - Toxicity prior Beta(0.15, 0.85)	6.9	5.2	5.5	6	5.6	
U2 - Toxicity prior Beta(0.3, 0.7)	6.6	4.8	5.2	6.4	6.2	
U3 - Toxicity prior Beta(0.3, 0.7)	6.5	4.8	5.2	6.4	6.2	
U2 - Toxicity prior Beta(0.5, 0.5)	6.3	4.5	5	6.7	6.6	
U3 - Toxicity prior Beta(0.5, 0.5)	6.3	4.5	5.1	6.7	6.6	
U2 - Toxicity prior Beta(1, 1)	6.2	4.4	5	6.6	7	
U3 - Toxicity prior Beta(1, 1)	6.2	4.4	4.9	6.6	7	
Mean IPDE patients						
U2 - Toxicity prior Beta(0.15, 0.85)	0.6	3.1	1.4	2	0.8	
U3 - Toxicity prior Beta(0.15, 0.85)	0.6	3	1.4	2	0.8	
U2 - Toxicity prior Beta(0.3, 0.7)	0.5	2.9	1.4	2.1	0.9	
U3 - Toxicity prior Beta(0.3, 0.7)	0.5	2.9	1.4	2	0.8	
U2 - Toxicity prior Beta(0.5, 0.5)	0.5	2.8	1.4	2	0.9	
U3 - Toxicity prior Beta(0.5, 0.5)	0.5	2.8	1.4	2	0.8	
U2 - Toxicity prior Beta(1, 1)	0.5	2.8	1.3	2	0.8	
U3 - Toxicity prior Beta(1, 1)	0.5	2.7	1.3	2	0.8	

Phase I/II Operating Characteristics

Random model - Toxicity prior comparison - allocation = two_stage - IPDE alpha = 0.3 - both utilities - N = 30

	Dose 1	Dose 2	Dose 3	Dose 4	Dose 5	Stop %
Scenario 15						
DLT rate	0.13	0.2	0.32	0.39	0.46	
Efficacy rate	0.1	0.1	0.1	0.1	0.1	
Utility 2	0	-0.1	-0.2	-0.3	-0.4	
Utility 3	40.8	38	33.2	30.4	27.6	
OBD selection (%)						
U2 - Toxicity prior Beta(0.15, 0.85)	52.1	26.1	10.1	3	1.7	7
U3 - Toxicity prior Beta(0.15, 0.85)	62.6	22.4	5.9	1.3	0.8	7
U2 - Toxicity prior Beta(0.3, 0.7)	53.6	24	10.3	3.7	1.4	7
U3 - Toxicity prior Beta(0.3, 0.7)	63.6	21.5	5.4	1.8	0.6	7.1
U2 - Toxicity prior Beta(0.5, 0.5)	51.8	25.4	11	3.3	1.6	6.9
U3 - Toxicity prior Beta(0.5, 0.5)	62.2	22.5	6	1.6	0.8	6.9
U2 - Toxicity prior Beta(1, 1)	53.4	24.1	10.7	3.5	1.3	7
U3 - Toxicity prior Beta(1, 1)	64.2	21.5	5.2	1.4	0.7	7
Mean patients treated						
U2 - Toxicity prior Beta(0.15, 0.85)	11.1	7	5.7	3.4	1.9	
U3 - Toxicity prior Beta(0.15, 0.85)	11.2	7	5.6	3.4	1.9	
U2 - Toxicity prior Beta(0.3, 0.7)	10.8	6.5	5.8	4	2.1	
U3 - Toxicity prior Beta(0.3, 0.7)	10.8	6.5	5.8	4	2.2	
U2 - Toxicity prior Beta(0.5, 0.5)	10.6	6.4	5.7	4.2	2.4	
U3 - Toxicity prior Beta(0.5, 0.5)	10.6	6.4	5.6	4.3	2.3	
U2 - Toxicity prior Beta(1, 1)	10.5	6.3	5.6	4.2	2.7	
U3 - Toxicity prior Beta(1, 1)	10.4	6.3	5.6	4.3	2.7	
Mean IPDE patients						
U2 - Toxicity prior Beta(0.15, 0.85)	1.2	3.4	1.5	1.4	0.5	
U3 - Toxicity prior Beta(0.15, 0.85)	1.1	3.4	1.5	1.4	0.4	
U2 - Toxicity prior Beta(0.3, 0.7)	1.1	3.2	1.6	1.5	0.5	
U3 - Toxicity prior Beta(0.3, 0.7)	1.2	3.2	1.6	1.5	0.5	
U2 - Toxicity prior Beta(0.5, 0.5)	1	3.1	1.5	1.5	0.5	
U3 - Toxicity prior Beta(0.5, 0.5)	1	3.1	1.5	1.5	0.5	
U2 - Toxicity prior Beta(1, 1)	1.1	3	1.3	1.4	0.4	
U3 - Toxicity prior Beta(1, 1)	1.1	3	1.4	1.4	0.4	
Scenario 16						
DLT rate	0.02	0.06	0.09	0.14	0.26	
Efficacy rate	0.2	0.2	0.2	0.2	0.2	
Utility 2	0.2	0.1	0.1	0.1	-0.1	
Utility 3	51.2	49.6	48.4	46.4	41.6	
OBD selection (%)						
U2 - Toxicity prior Beta(0.15, 0.85)	30.3	24.3	18.2	15.3	11.8	0.1
U3 - Toxicity prior Beta(0.15, 0.85)	39.4	28.2	15.2	11.2	5.9	0.1
U2 - Toxicity prior Beta(0.3, 0.7)	30.5	25	16.3	15.4	12.7	0.1
U3 - Toxicity prior Beta(0.3, 0.7)	37.2	29.4	16.1	11.2	6	0.1
U2 - Toxicity prior Beta(0.5, 0.5)	29.1	24.9	17.1	16.2	12.6	0.1
U3 - Toxicity prior Beta(0.5, 0.5)	35.1	28.6	17.2	12.6	6.4	0.1
U2 - Toxicity prior Beta(1, 1)	28.6	24.4	18.2	16	12.7	0.1
U3 - Toxicity prior Beta(1, 1)	34.8	29.6	16.2	12.6	6.7	0.1
Mean patients treated						
U2 - Toxicity prior Beta(0.15, 0.85)	6	5.5	5.4	5.8	7.4	
U3 - Toxicity prior Beta(0.15, 0.85)	6.1	5.4	5.4	5.7	7.4	
U2 - Toxicity prior Beta(0.3, 0.7)	6	5.4	5.3	5.7	7.5	
U3 - Toxicity prior Beta(0.3, 0.7)	6	5.4	5.4	5.7	7.5	
U2 - Toxicity prior Beta(0.5, 0.5)	6	5.4	5.3	5.7	7.6	
U3 - Toxicity prior Beta(0.5, 0.5)	5.9	5.4	5.3	5.8	7.6	
U2 - Toxicity prior Beta(1, 1)	6	5.4	5.3	5.7	7.6	
U3 - Toxicity prior Beta(1, 1)	6	5.4	5.2	5.7	7.6	
Mean IPDE patients						
U2 - Toxicity prior Beta(0.15, 0.85)	1.1	3.2	1.8	2.6	1.5	
U3 - Toxicity prior Beta(0.15, 0.85)	1.1	3.2	1.8	2.6	1.5	
U2 - Toxicity prior Beta(0.3, 0.7)	1.1	3.2	1.8	2.6	1.5	
U3 - Toxicity prior Beta(0.3, 0.7)	1.1	3.1	1.8	2.6	1.5	
U2 - Toxicity prior Beta(0.5, 0.5)	1.1	3.1	1.8	2.6	1.5	
U3 - Toxicity prior Beta(0.5, 0.5)	1.1	3.1	1.8	2.6	1.5	
U2 - Toxicity prior Beta(1, 1)	1.1	3.1	1.8	2.6	1.5	
U3 - Toxicity prior Beta(1, 1)	1.1	3.1	1.8	2.6	1.5	

Phase I/II Operating Characteristics

Random model - Toxicity prior comparison - allocation = two_stage - IPDE alpha = 0.3 - both utilities - N = 30

	Dose 1	Dose 2	Dose 3	Dose 4	Dose 5	Stop %
Scenario 17						
DLT rate	0.11	0.23	0.39	0.58	0.76	
Efficacy rate	0.3	0.5	0.5	0.5	0.5	
Utility 2	0.2	0.3	0.1	-0.1	-0.3	
Utility 3	53.6	60.8	54.4	46.8	39.6	
OBD selection (%)						
U2 - Toxicity prior Beta(0.15, 0.85)	32.7	48.1	17.2	1.7	0	0.3
U3 - Toxicity prior Beta(0.15, 0.85)	34.7	49.1	14.6	1.3	0	0.3
U2 - Toxicity prior Beta(0.3, 0.7)	32.5	47.9	17.6	1.7	0	0.3
U3 - Toxicity prior Beta(0.3, 0.7)	35.3	47.6	15.6	1.2	0	0.3
U2 - Toxicity prior Beta(0.5, 0.5)	31.6	47.8	17.7	2.6	0	0.3
U3 - Toxicity prior Beta(0.5, 0.5)	34.4	48.5	15.2	1.6	0	0.3
U2 - Toxicity prior Beta(1, 1)	31	47.4	18.2	3.1	0	0.3
U3 - Toxicity prior Beta(1, 1)	33.6	47.6	16.1	2.4	0	0.3
Mean patients treated						
U2 - Toxicity prior Beta(0.15, 0.85)	9.9	9.7	7.2	2.6	0.5	
U3 - Toxicity prior Beta(0.15, 0.85)	9.9	9.8	7.2	2.5	0.5	
U2 - Toxicity prior Beta(0.3, 0.7)	9.5	9.6	7.3	3	0.5	
U3 - Toxicity prior Beta(0.3, 0.7)	9.4	9.6	7.4	3	0.5	
U2 - Toxicity prior Beta(0.5, 0.5)	9.3	9.4	7.3	3.3	0.6	
U3 - Toxicity prior Beta(0.5, 0.5)	9.2	9.4	7.4	3.3	0.6	
U2 - Toxicity prior Beta(1, 1)	9.1	9.5	7.4	3.3	0.7	
U3 - Toxicity prior Beta(1, 1)	9.2	9.5	7.3	3.3	0.7	
Mean IPDE patients						
U2 - Toxicity prior Beta(0.15, 0.85)	0.4	2.8	1.1	0.6	0.1	
U3 - Toxicity prior Beta(0.15, 0.85)	0.5	2.8	1.1	0.6	0.1	
U2 - Toxicity prior Beta(0.3, 0.7)	0.4	2.6	1.2	0.7	0.1	
U3 - Toxicity prior Beta(0.3, 0.7)	0.5	2.7	1.2	0.7	0.1	
U2 - Toxicity prior Beta(0.5, 0.5)	0.4	2.6	1.1	0.6	0.1	
U3 - Toxicity prior Beta(0.5, 0.5)	0.4	2.6	1.1	0.6	0.1	
U2 - Toxicity prior Beta(1, 1)	0.4	2.5	1	0.6	0.1	
U3 - Toxicity prior Beta(1, 1)	0.4	2.5	1	0.6	0.1	
Scenario 18						
DLT rate	0.02	0.04	0.1	0.2	0.35	
Efficacy rate	0.05	0.1	0.15	0.2	0.5	
Utility 2	0	0.1	0.1	0	0.1	
Utility 3	42.2	44.4	45	44	56	
OBD selection (%)						
U2 - Toxicity prior Beta(0.15, 0.85)	9.7	8.4	11.2	14.3	55.7	0.7
U3 - Toxicity prior Beta(0.15, 0.85)	16.9	11.4	10	12.9	48.1	0.7
U2 - Toxicity prior Beta(0.3, 0.7)	10.2	8.7	11.2	13.8	55.4	0.7
U3 - Toxicity prior Beta(0.3, 0.7)	15.7	13.4	9.4	12.2	48.6	0.7
U2 - Toxicity prior Beta(0.5, 0.5)	10.8	7.2	11.7	13.4	56.2	0.7
U3 - Toxicity prior Beta(0.5, 0.5)	16.4	11.8	9.8	12	49.3	0.7
U2 - Toxicity prior Beta(1, 1)	9.8	7.1	10.8	13	58.6	0.7
U3 - Toxicity prior Beta(1, 1)	14	11.9	9.5	11	52.9	0.7
Mean patients treated						
U2 - Toxicity prior Beta(0.15, 0.85)	3.8	3.8	4.7	5.4	12.3	
U3 - Toxicity prior Beta(0.15, 0.85)	3.8	3.7	4.6	5.5	12.3	
U2 - Toxicity prior Beta(0.3, 0.7)	3.8	3.7	4.6	5.4	12.5	
U3 - Toxicity prior Beta(0.3, 0.7)	3.8	3.7	4.5	5.4	12.5	
U2 - Toxicity prior Beta(0.5, 0.5)	3.8	3.7	4.5	5.4	12.6	
U3 - Toxicity prior Beta(0.5, 0.5)	3.8	3.7	4.5	5.4	12.6	
U2 - Toxicity prior Beta(1, 1)	3.7	3.6	4.4	5.3	12.9	
U3 - Toxicity prior Beta(1, 1)	3.7	3.6	4.4	5.3	12.9	
Mean IPDE patients						
U2 - Toxicity prior Beta(0.15, 0.85)	0.2	2.7	1.4	2.3	1.4	
U3 - Toxicity prior Beta(0.15, 0.85)	0.2	2.7	1.4	2.3	1.5	
U2 - Toxicity prior Beta(0.3, 0.7)	0.2	2.7	1.4	2.3	1.4	
U3 - Toxicity prior Beta(0.3, 0.7)	0.2	2.7	1.4	2.3	1.4	
U2 - Toxicity prior Beta(0.5, 0.5)	0.2	2.7	1.4	2.3	1.4	
U3 - Toxicity prior Beta(0.5, 0.5)	0.2	2.7	1.4	2.3	1.4	
U2 - Toxicity prior Beta(1, 1)	0.2	2.7	1.4	2.3	1.4	
U3 - Toxicity prior Beta(1, 1)	0.2	2.7	1.4	2.3	1.4	

Phase I/II Operating Characteristics

Random model - Toxicity prior comparison - allocation = two_stage - IPDE alpha = 0.3 - both utilities - N = 30

	Dose 1	Dose 2	Dose 3	Dose 4	Dose 5	Stop %
Scenario 19						
DLT rate	0.11	0.26	0.39	0.6	0.77	
Efficacy rate	0.08	0.3	0.32	0.35	0.4	
Utility 2	0	0	-0.1	-0.2	-0.4	
Utility 3	40.4	47.6	43.6	37	33.2	
OBD selection (%)						
U2 - Toxicity prior Beta(0.15, 0.85)	22.8	50.3	16.6	2	0	8.3
U3 - Toxicity prior Beta(0.15, 0.85)	26.1	50.7	13	1.7	0	8.5
U2 - Toxicity prior Beta(0.3, 0.7)	23	48.5	17.4	2.7	0	8.4
U3 - Toxicity prior Beta(0.3, 0.7)	26	49.5	14.6	1.7	0	8.2
U2 - Toxicity prior Beta(0.5, 0.5)	23.6	49.5	16.4	2.3	0	8.2
U3 - Toxicity prior Beta(0.5, 0.5)	26.3	47.9	16	1.6	0	8.2
U2 - Toxicity prior Beta(1, 1)	23.1	48.2	17.5	3.2	0	8
U3 - Toxicity prior Beta(1, 1)	25.7	48.9	15	2.3	0	8.1
Mean patients treated						
U2 - Toxicity prior Beta(0.15, 0.85)	8.9	10	7.2	2.5	0.5	
U3 - Toxicity prior Beta(0.15, 0.85)	8.8	10.2	7.1	2.4	0.5	
U2 - Toxicity prior Beta(0.3, 0.7)	8.5	9.7	7.4	3	0.6	
U3 - Toxicity prior Beta(0.3, 0.7)	8.6	9.7	7.3	3	0.6	
U2 - Toxicity prior Beta(0.5, 0.5)	8.4	9.3	7.4	3.4	0.7	
U3 - Toxicity prior Beta(0.5, 0.5)	8.3	9.4	7.4	3.4	0.7	
U2 - Toxicity prior Beta(1, 1)	8.3	9.3	7.3	3.5	0.8	
U3 - Toxicity prior Beta(1, 1)	8.3	9.4	7.2	3.5	0.8	
Mean IPDE patients						
U2 - Toxicity prior Beta(0.15, 0.85)	0.5	3.5	1.3	0.8	0.1	
U3 - Toxicity prior Beta(0.15, 0.85)	0.4	3.5	1.3	0.8	0.1	
U2 - Toxicity prior Beta(0.3, 0.7)	0.5	3.3	1.3	0.9	0.1	
U3 - Toxicity prior Beta(0.3, 0.7)	0.5	3.3	1.3	0.8	0.1	
U2 - Toxicity prior Beta(0.5, 0.5)	0.5	3.3	1.2	0.8	0.1	
U3 - Toxicity prior Beta(0.5, 0.5)	0.4	3.2	1.2	0.8	0.1	
U2 - Toxicity prior Beta(1, 1)	0.4	3	1.1	0.7	0.1	
U3 - Toxicity prior Beta(1, 1)	0.4	3	1.1	0.7	0.1	
Scenario 20						
DLT rate	0.07	0.14	0.31	0.41	0.52	
Efficacy rate	0.05	0.25	0.25	0.25	0.25	
Utility 2	0	0.1	-0.1	-0.2	-0.3	
Utility 3	40.2	49.4	42.6	38.6	34.2	
OBD selection (%)						
U2 - Toxicity prior Beta(0.15, 0.85)	14	43.4	28.2	8.7	1.4	4.3
U3 - Toxicity prior Beta(0.15, 0.85)	18.9	46.5	22.5	7.1	0.7	4.3
U2 - Toxicity prior Beta(0.3, 0.7)	13.4	42.6	27.7	10.5	1.6	4.2
U3 - Toxicity prior Beta(0.3, 0.7)	20.6	44.8	21.9	7.6	0.9	4.2
U2 - Toxicity prior Beta(0.5, 0.5)	13	41.3	29.5	9.9	2.1	4.2
U3 - Toxicity prior Beta(0.5, 0.5)	21.7	44.4	21.7	7.2	0.8	4.2
U2 - Toxicity prior Beta(1, 1)	13.9	38.6	29	11.5	2.8	4.2
U3 - Toxicity prior Beta(1, 1)	20	43.2	23.4	7.9	1.3	4.2
Mean patients treated						
U2 - Toxicity prior Beta(0.15, 0.85)	5.7	8.3	8.3	4.6	2.5	
U3 - Toxicity prior Beta(0.15, 0.85)	5.7	8.3	8	4.9	2.5	
U2 - Toxicity prior Beta(0.3, 0.7)	5.5	7.8	8	5.4	2.7	
U3 - Toxicity prior Beta(0.3, 0.7)	5.5	7.8	8	5.4	2.8	
U2 - Toxicity prior Beta(0.5, 0.5)	5.5	7.5	7.9	5.7	2.9	
U3 - Toxicity prior Beta(0.5, 0.5)	5.5	7.5	7.8	5.7	3	
U2 - Toxicity prior Beta(1, 1)	5.4	7.3	7.7	5.8	3.3	
U3 - Toxicity prior Beta(1, 1)	5.4	7.4	7.6	5.8	3.3	
Mean IPDE patients						
U2 - Toxicity prior Beta(0.15, 0.85)	0.5	3.5	1.8	1.6	0.5	
U3 - Toxicity prior Beta(0.15, 0.85)	0.4	3.5	1.8	1.6	0.5	
U2 - Toxicity prior Beta(0.3, 0.7)	0.4	3.4	1.8	1.7	0.5	
U3 - Toxicity prior Beta(0.3, 0.7)	0.4	3.4	1.8	1.6	0.6	
U2 - Toxicity prior Beta(0.5, 0.5)	0.4	3.4	1.7	1.7	0.5	
U3 - Toxicity prior Beta(0.5, 0.5)	0.4	3.4	1.8	1.6	0.5	
U2 - Toxicity prior Beta(1, 1)	0.4	3.3	1.6	1.6	0.5	
U3 - Toxicity prior Beta(1, 1)	0.4	3.3	1.6	1.6	0.5	

Phase I/II Operating Characteristics

Random model - Toxicity prior comparison - allocation = two_stage - IPDE alpha = 0.3 - both utilities - N = 30

	Dose 1	Dose 2	Dose 3	Dose 4	Dose 5	Stop %
Scenario 21						
DLT rate	0.37	0.56	0.69	0.83	0.88	
Efficacy rate	0.5	0.5	0.5	0.5	0.5	
Utility 2	0.1	-0.1	-0.2	-0.3	-0.4	
Utility 3	55.2	47.6	42.4	36.8	34.8	
OBD selection (%)						
U2 - Toxicity prior Beta(0.15, 0.85)	59	1	0	0	0	40
U3 - Toxicity prior Beta(0.15, 0.85)	59.1	1.2	0	0	0	39.7
U2 - Toxicity prior Beta(0.3, 0.7)	59.5	1.3	0	0	0	39.2
U3 - Toxicity prior Beta(0.3, 0.7)	59.9	1.5	0	0	0	38.6
U2 - Toxicity prior Beta(0.5, 0.5)	60.2	1	0.1	0	0	38.7
U3 - Toxicity prior Beta(0.5, 0.5)	60.4	1	0	0	0	38.6
U2 - Toxicity prior Beta(1, 1)	60.8	1	0.1	0	0	38.1
U3 - Toxicity prior Beta(1, 1)	60.4	1.2	0	0	0	38.4
Mean patients treated						
U2 - Toxicity prior Beta(0.15, 0.85)	20.5	2.3	0.6	0.1	0	
U3 - Toxicity prior Beta(0.15, 0.85)	20.6	2.2	0.6	0.1	0	
U2 - Toxicity prior Beta(0.3, 0.7)	20.4	2.4	0.7	0.1	0	
U3 - Toxicity prior Beta(0.3, 0.7)	20.3	2.5	0.7	0.1	0	
U2 - Toxicity prior Beta(0.5, 0.5)	20.3	2.5	0.7	0.2	0	
U3 - Toxicity prior Beta(0.5, 0.5)	20.3	2.5	0.7	0.2	0	
U2 - Toxicity prior Beta(1, 1)	20.3	2.6	0.8	0.2	0	
U3 - Toxicity prior Beta(1, 1)	20.2	2.6	0.8	0.2	0	
Mean IPDE patients						
U2 - Toxicity prior Beta(0.15, 0.85)	0.1	0.7	0.1	0	0	
U3 - Toxicity prior Beta(0.15, 0.85)	0.2	0.7	0.1	0	0	
U2 - Toxicity prior Beta(0.3, 0.7)	0.2	0.6	0.1	0	0	
U3 - Toxicity prior Beta(0.3, 0.7)	0.2	0.7	0.1	0	0	
U2 - Toxicity prior Beta(0.5, 0.5)	0.1	0.6	0.1	0	0	
U3 - Toxicity prior Beta(0.5, 0.5)	0.1	0.6	0.1	0	0	
U2 - Toxicity prior Beta(1, 1)	0.1	0.6	0.1	0	0	
U3 - Toxicity prior Beta(1, 1)	0.1	0.6	0.1	0	0	
Scenario 22						
DLT rate	0.07	0.27	0.5	0.56	0.78	
Efficacy rate	0.08	0.25	0.25	0.5	0.5	
Utility 2	0	0	-0.2	-0.1	-0.3	
Utility 3	42	44.2	35	47.6	38.8	
OBD selection (%)						
U2 - Toxicity prior Beta(0.15, 0.85)	28	52.2	13.5	1.7	0	4.6
U3 - Toxicity prior Beta(0.15, 0.85)	32.7	50.9	10.4	1.4	0	4.6
U2 - Toxicity prior Beta(0.3, 0.7)	29.4	50.3	13.3	2.2	0.1	4.7
U3 - Toxicity prior Beta(0.3, 0.7)	32.9	49.9	10.9	1.7	0	4.6
U2 - Toxicity prior Beta(0.5, 0.5)	31.4	47.3	14.6	2.4	0	4.3
U3 - Toxicity prior Beta(0.5, 0.5)	33.7	48.2	11.4	2.4	0	4.3
U2 - Toxicity prior Beta(1, 1)	28.3	50.7	13.7	2.9	0.1	4.3
U3 - Toxicity prior Beta(1, 1)	34.8	48.8	10	2.2	0	4.2
Mean patients treated						
U2 - Toxicity prior Beta(0.15, 0.85)	8.8	11.9	6.5	1.9	0.4	
U3 - Toxicity prior Beta(0.15, 0.85)	8.8	11.8	6.4	2	0.4	
U2 - Toxicity prior Beta(0.3, 0.7)	8.4	11.3	6.7	2.6	0.5	
U3 - Toxicity prior Beta(0.3, 0.7)	8.4	11.4	6.6	2.6	0.5	
U2 - Toxicity prior Beta(0.5, 0.5)	8.3	10.8	6.8	3	0.6	
U3 - Toxicity prior Beta(0.5, 0.5)	8.3	10.8	6.8	3	0.6	
U2 - Toxicity prior Beta(1, 1)	8	10.9	6.8	3.2	0.7	
U3 - Toxicity prior Beta(1, 1)	8.1	10.8	6.8	3.1	0.7	
Mean IPDE patients						
U2 - Toxicity prior Beta(0.15, 0.85)	0.8	3.9	1.2	0.5	0.1	
U3 - Toxicity prior Beta(0.15, 0.85)	0.7	3.8	1.2	0.5	0.1	
U2 - Toxicity prior Beta(0.3, 0.7)	0.8	3.6	1.2	0.7	0.1	
U3 - Toxicity prior Beta(0.3, 0.7)	0.8	3.6	1.2	0.7	0.1	
U2 - Toxicity prior Beta(0.5, 0.5)	0.7	3.5	1.1	0.6	0.1	
U3 - Toxicity prior Beta(0.5, 0.5)	0.7	3.5	1.1	0.6	0.1	
U2 - Toxicity prior Beta(1, 1)	0.7	3.3	1	0.6	0.1	
U3 - Toxicity prior Beta(1, 1)	0.7	3.3	1	0.6	0.1	

Phase I/II Operating Characteristics

Random model - Toxicity prior comparison - allocation = two_stage - IPDE alpha = 0.3 - both utilities - N = 30

	Dose 1	Dose 2	Dose 3	Dose 4	Dose 5	Stop %
Scenario 23						
DLT rate	0.32	0.55	0.8	0.94	0.97	
Efficacy rate	0.05	0.05	0.05	0.05	0.05	
Utility 2	-0.3	-0.5	-0.8	-0.9	-0.9	
Utility 3	30.2	21	11	5.4	4.2	
OBD selection (%)						
U2 - Toxicity prior Beta(0.15, 0.85)	21.6	4.1	0.4	0	0	73.9
U3 - Toxicity prior Beta(0.15, 0.85)	21.2	3.8	0.1	0	0	74.9
U2 - Toxicity prior Beta(0.3, 0.7)	22.3	3.4	0.4	0	0	73.9
U3 - Toxicity prior Beta(0.3, 0.7)	21.8	3.5	0	0	0	74.7
U2 - Toxicity prior Beta(0.5, 0.5)	21.3	4.2	0.1	0	0	74.4
U3 - Toxicity prior Beta(0.5, 0.5)	21.8	3.8	0.1	0	0	74.3
U2 - Toxicity prior Beta(1, 1)	21.2	3.5	0.4	0	0	74.9
U3 - Toxicity prior Beta(1, 1)	22.1	3.5	0	0	0	74.4
Mean patients treated						
U2 - Toxicity prior Beta(0.15, 0.85)	16.6	3.2	1	0.1	0	
U3 - Toxicity prior Beta(0.15, 0.85)	16.6	3.1	1	0.1	0	
U2 - Toxicity prior Beta(0.3, 0.7)	16.6	3.3	1.1	0.1	0	
U3 - Toxicity prior Beta(0.3, 0.7)	16.6	3.3	1.1	0.1	0	
U2 - Toxicity prior Beta(0.5, 0.5)	16.7	3.2	1.2	0.1	0	
U3 - Toxicity prior Beta(0.5, 0.5)	16.7	3.1	1.2	0.1	0	
U2 - Toxicity prior Beta(1, 1)	16.6	3.3	1.2	0.1	0	
U3 - Toxicity prior Beta(1, 1)	16.6	3.3	1.2	0.1	0	
Mean IPDE patients						
U2 - Toxicity prior Beta(0.15, 0.85)	0.4	1.5	0.2	0	0	
U3 - Toxicity prior Beta(0.15, 0.85)	0.4	1.5	0.2	0	0	
U2 - Toxicity prior Beta(0.3, 0.7)	0.4	1.4	0.2	0	0	
U3 - Toxicity prior Beta(0.3, 0.7)	0.4	1.4	0.2	0	0	
U2 - Toxicity prior Beta(0.5, 0.5)	0.3	1.3	0.2	0	0	
U3 - Toxicity prior Beta(0.5, 0.5)	0.3	1.3	0.2	0	0	
U2 - Toxicity prior Beta(1, 1)	0.1	1.1	0.2	0	0	
U3 - Toxicity prior Beta(1, 1)	0.2	1.2	0.2	0	0	
Scenario 24						
DLT rate	0.01	0.02	0.05	0.1	0.25	
Efficacy rate	0.05	0.08	0.2	0.5	0.5	
Utility 2	0	0.1	0.1	0.4	0.2	
Utility 3	42.6	44	50	66	60	
OBD selection (%)						
U2 - Toxicity prior Beta(0.15, 0.85)	1.6	1.1	6.7	42.7	47.8	0.1
U3 - Toxicity prior Beta(0.15, 0.85)	2.7	2.4	7.6	45	42.2	0.1
U2 - Toxicity prior Beta(0.3, 0.7)	1.4	0.9	5.9	43.5	48.3	0
U3 - Toxicity prior Beta(0.3, 0.7)	2.8	1.9	7.4	45.3	42.6	0
U2 - Toxicity prior Beta(0.5, 0.5)	1.6	1.4	6.4	43.1	47.5	0
U3 - Toxicity prior Beta(0.5, 0.5)	2.9	2.6	7.1	45.4	42	0
U2 - Toxicity prior Beta(1, 1)	1	1.4	5.9	44	47.7	0
U3 - Toxicity prior Beta(1, 1)	2.8	2	6.8	45.7	42.7	0
Mean patients treated						
U2 - Toxicity prior Beta(0.15, 0.85)	3.3	3.2	3.7	8.2	11.7	
U3 - Toxicity prior Beta(0.15, 0.85)	3.3	3.1	3.7	8.2	11.7	
U2 - Toxicity prior Beta(0.3, 0.7)	3.3	3.1	3.7	8.2	11.8	
U3 - Toxicity prior Beta(0.3, 0.7)	3.2	3.1	3.6	8.2	11.8	
U2 - Toxicity prior Beta(0.5, 0.5)	3.3	3.1	3.7	8.1	11.8	
U3 - Toxicity prior Beta(0.5, 0.5)	3.3	3.1	3.7	8.2	11.8	
U2 - Toxicity prior Beta(1, 1)	3.3	3.1	3.7	8.2	11.8	
U3 - Toxicity prior Beta(1, 1)	3.3	3.1	3.7	8.2	11.8	
Mean IPDE patients						
U2 - Toxicity prior Beta(0.15, 0.85)	0.1	2.5	1.2	3.1	1.1	
U3 - Toxicity prior Beta(0.15, 0.85)	0.1	2.5	1.2	3.1	1.1	
U2 - Toxicity prior Beta(0.3, 0.7)	0.1	2.4	1.2	3.1	1.2	
U3 - Toxicity prior Beta(0.3, 0.7)	0.1	2.5	1.2	3.1	1.2	
U2 - Toxicity prior Beta(0.5, 0.5)	0.1	2.5	1.2	3.1	1.2	
U3 - Toxicity prior Beta(0.5, 0.5)	0.1	2.5	1.2	3.1	1.1	
U2 - Toxicity prior Beta(1, 1)	0.1	2.5	1.2	3.1	1.1	
U3 - Toxicity prior Beta(1, 1)	0.1	2.5	1.2	3.1	1.1	

Phase I/II Operating Characteristics

Random model - Toxicity prior comparison - allocation = two_stage - IPDE alpha = 0.3 - both utilities - N = 30

	Dose 1	Dose 2	Dose 3	Dose 4	Dose 5	Stop %
Scenario 25						
DLT rate	0.09	0.18	0.37	0.63	0.77	
Efficacy rate	0.2	0.25	0.4	0.45	0.5	
Utility 2	0.1	0.1	0	-0.2	-0.3	
Utility 3	48.4	47.8	49.2	41.8	39.2	
OBD selection (%)						
U2 - Toxicity prior Beta(0.15, 0.85)	36.5	28.7	29.6	4.5	0	0.7
U3 - Toxicity prior Beta(0.15, 0.85)	40.3	30.5	24.8	3.7	0	0.7
U2 - Toxicity prior Beta(0.3, 0.7)	36	28.6	30.2	4.5	0	0.7
U3 - Toxicity prior Beta(0.3, 0.7)	41.7	27.4	26.2	4	0	0.7
U2 - Toxicity prior Beta(0.5, 0.5)	35.7	27.5	31.2	4.8	0.1	0.7
U3 - Toxicity prior Beta(0.5, 0.5)	41.1	27.8	26.4	4	0	0.7
U2 - Toxicity prior Beta(1, 1)	36.9	26.9	29.7	5.7	0.1	0.7
U3 - Toxicity prior Beta(1, 1)	41.5	27.9	25.6	4.2	0.1	0.7
Mean patients treated						
U2 - Toxicity prior Beta(0.15, 0.85)	9.4	8.2	8.8	3	0.6	
U3 - Toxicity prior Beta(0.15, 0.85)	9.3	8	9.1	2.9	0.6	
U2 - Toxicity prior Beta(0.3, 0.7)	9	7.6	9	3.8	0.7	
U3 - Toxicity prior Beta(0.3, 0.7)	9	7.6	8.9	3.8	0.7	
U2 - Toxicity prior Beta(0.5, 0.5)	8.8	7.4	8.9	4	0.7	
U3 - Toxicity prior Beta(0.5, 0.5)	8.9	7.3	8.9	4	0.8	
U2 - Toxicity prior Beta(1, 1)	8.7	7.3	8.9	4.1	0.9	
U3 - Toxicity prior Beta(1, 1)	8.7	7.3	9	4	0.9	
Mean IPDE patients						
U2 - Toxicity prior Beta(0.15, 0.85)	0.8	3.1	1.5	0.9	0.1	
U3 - Toxicity prior Beta(0.15, 0.85)	0.7	3	1.5	0.9	0.1	
U2 - Toxicity prior Beta(0.3, 0.7)	0.7	2.9	1.5	0.9	0.1	
U3 - Toxicity prior Beta(0.3, 0.7)	0.7	2.9	1.5	0.9	0.1	
U2 - Toxicity prior Beta(0.5, 0.5)	0.8	2.9	1.4	0.9	0.1	
U3 - Toxicity prior Beta(0.5, 0.5)	0.7	2.8	1.4	0.9	0.1	
U2 - Toxicity prior Beta(1, 1)	0.8	2.7	1.3	0.8	0.1	
U3 - Toxicity prior Beta(1, 1)	0.7	2.7	1.3	0.8	0.1	
Scenario 26						
DLT rate	0.15	0.22	0.3	0.62	0.8	
Efficacy rate	0.1	0.2	0.35	0.45	0.55	
Utility 2	-0.1	0	0.1	-0.2	-0.2	
Utility 3	40	43.2	49	42.2	41	
OBD selection (%)						
U2 - Toxicity prior Beta(0.15, 0.85)	33.3	26.9	24.8	4.8	0.1	10.1
U3 - Toxicity prior Beta(0.15, 0.85)	36.1	26.9	21.9	4.8	0.1	10.2
U2 - Toxicity prior Beta(0.3, 0.7)	31.6	25.5	26.8	6.3	0	9.8
U3 - Toxicity prior Beta(0.3, 0.7)	35.7	25.1	23.4	6	0	9.8
U2 - Toxicity prior Beta(0.5, 0.5)	30.9	25.3	26.4	7.6	0.1	9.7
U3 - Toxicity prior Beta(0.5, 0.5)	35.1	25	24.1	6.1	0	9.7
U2 - Toxicity prior Beta(1, 1)	31.6	25.6	25.4	7.6	0.1	9.7
U3 - Toxicity prior Beta(1, 1)	35.8	24.4	23	7	0	9.8
Mean patients treated						
U2 - Toxicity prior Beta(0.15, 0.85)	10.8	6.8	7.8	3.1	0.6	
U3 - Toxicity prior Beta(0.15, 0.85)	10.8	6.7	7.9	3	0.6	
U2 - Toxicity prior Beta(0.3, 0.7)	10.4	6.5	7.9	3.7	0.6	
U3 - Toxicity prior Beta(0.3, 0.7)	10.3	6.4	7.9	3.7	0.6	
U2 - Toxicity prior Beta(0.5, 0.5)	10.2	6.2	8	4	0.7	
U3 - Toxicity prior Beta(0.5, 0.5)	10.2	6.2	8	3.9	0.7	
U2 - Toxicity prior Beta(1, 1)	10.1	6.1	7.9	4	0.8	
U3 - Toxicity prior Beta(1, 1)	10.1	6.1	7.8	4.1	0.8	
Mean IPDE patients						
U2 - Toxicity prior Beta(0.15, 0.85)	0.5	3	1.2	0.9	0.1	
U3 - Toxicity prior Beta(0.15, 0.85)	0.5	3	1.3	0.9	0.1	
U2 - Toxicity prior Beta(0.3, 0.7)	0.5	2.8	1.3	1	0.1	
U3 - Toxicity prior Beta(0.3, 0.7)	0.6	2.8	1.3	1	0.1	
U2 - Toxicity prior Beta(0.5, 0.5)	0.5	2.7	1.2	0.9	0.1	
U3 - Toxicity prior Beta(0.5, 0.5)	0.5	2.7	1.1	0.9	0.1	
U2 - Toxicity prior Beta(1, 1)	0.5	2.6	1	0.8	0.1	
U3 - Toxicity prior Beta(1, 1)	0.5	2.6	1.1	0.9	0.1	

Phase I/II Operating Characteristics

Random model - Toxicity prior comparison - allocation = two_stage - IPDE alpha = 0.3 - both utilities - N = 30

	Dose 1	Dose 2	Dose 3	Dose 4	Dose 5	Stop %
Scenario 27						
DLT rate	0.11	0.17	0.3	0.5	0.62	
Efficacy rate	0.1	0.2	0.4	0.42	0.45	
Utility 2	0	0	0.1	-0.1	-0.2	
Utility 3	41.6	45.2	52	45.2	42.2	
OBD selection (%)						
U2 - Toxicity prior Beta(0.15, 0.85)	23	24	40	6.7	0.9	5.4
U3 - Toxicity prior Beta(0.15, 0.85)	26.9	24.7	36.4	6.2	0.4	5.4
U2 - Toxicity prior Beta(0.3, 0.7)	21.7	23.2	40.6	8.2	0.9	5.4
U3 - Toxicity prior Beta(0.3, 0.7)	26.3	22.8	37.1	7.9	0.5	5.4
U2 - Toxicity prior Beta(0.5, 0.5)	21.1	21.7	41.7	8.9	1.3	5.3
U3 - Toxicity prior Beta(0.5, 0.5)	26	22	37.7	8.3	0.7	5.3
U2 - Toxicity prior Beta(1, 1)	22	20.2	40.7	10.3	1.4	5.4
U3 - Toxicity prior Beta(1, 1)	26.7	20.6	37.1	9.3	0.9	5.4
Mean patients treated						
U2 - Toxicity prior Beta(0.15, 0.85)	8.1	6.5	9.2	4.2	1.5	
U3 - Toxicity prior Beta(0.15, 0.85)	8.1	6.4	9.3	4.1	1.6	
U2 - Toxicity prior Beta(0.3, 0.7)	7.8	6.1	8.9	5	1.7	
U3 - Toxicity prior Beta(0.3, 0.7)	7.8	6	8.8	5	1.7	
U2 - Toxicity prior Beta(0.5, 0.5)	7.6	5.8	8.9	5.3	1.9	
U3 - Toxicity prior Beta(0.5, 0.5)	7.6	5.7	8.8	5.4	1.9	
U2 - Toxicity prior Beta(1, 1)	7.7	5.7	8.5	5.5	2.1	
U3 - Toxicity prior Beta(1, 1)	7.6	5.6	8.6	5.5	2.1	
Mean IPDE patients						
U2 - Toxicity prior Beta(0.15, 0.85)	0.5	2.9	1.6	1.1	0.3	
U3 - Toxicity prior Beta(0.15, 0.85)	0.4	2.9	1.7	1.1	0.3	
U2 - Toxicity prior Beta(0.3, 0.7)	0.4	2.8	1.7	1.2	0.3	
U3 - Toxicity prior Beta(0.3, 0.7)	0.4	2.7	1.6	1.2	0.3	
U2 - Toxicity prior Beta(0.5, 0.5)	0.4	2.7	1.6	1.1	0.3	
U3 - Toxicity prior Beta(0.5, 0.5)	0.4	2.7	1.6	1.1	0.3	
U2 - Toxicity prior Beta(1, 1)	0.4	2.7	1.4	1.1	0.3	
U3 - Toxicity prior Beta(1, 1)	0.4	2.7	1.5	1.1	0.3	
Scenario 28						
DLT rate	0.2	0.3	0.5	0.6	0.77	
Efficacy rate	0.05	0.3	0.32	0.35	0.4	
Utility 2	-0.1	0	-0.2	-0.2	-0.4	
Utility 3	35	46	39.2	37	33.2	
OBD selection (%)						
U2 - Toxicity prior Beta(0.15, 0.85)	25.7	34.6	6.9	0.1	0	32.7
U3 - Toxicity prior Beta(0.15, 0.85)	25.6	35.7	5.5	0.1	0	33.1
U2 - Toxicity prior Beta(0.3, 0.7)	25.5	34.5	7.4	0.3	0	32.3
U3 - Toxicity prior Beta(0.3, 0.7)	26.9	35.4	5.7	0.2	0	31.8
U2 - Toxicity prior Beta(0.5, 0.5)	26.1	33.2	8	0.5	0	32.2
U3 - Toxicity prior Beta(0.5, 0.5)	26.9	34.2	6.4	0.5	0	32
U2 - Toxicity prior Beta(1, 1)	25.8	33.9	7.4	0.8	0	32.1
U3 - Toxicity prior Beta(1, 1)	27.5	34.1	5.3	0.8	0	32.3
Mean patients treated						
U2 - Toxicity prior Beta(0.15, 0.85)	11.9	8.9	4.2	1	0.2	
U3 - Toxicity prior Beta(0.15, 0.85)	11.9	8.9	4.2	1	0.2	
U2 - Toxicity prior Beta(0.3, 0.7)	11.6	8.4	4.7	1.3	0.3	
U3 - Toxicity prior Beta(0.3, 0.7)	11.5	8.6	4.7	1.3	0.2	
U2 - Toxicity prior Beta(0.5, 0.5)	11.4	8.1	5	1.6	0.3	
U3 - Toxicity prior Beta(0.5, 0.5)	11.4	8.2	5	1.6	0.3	
U2 - Toxicity prior Beta(1, 1)	11.3	8.2	5	1.7	0.3	
U3 - Toxicity prior Beta(1, 1)	11.3	8.2	4.9	1.7	0.3	
Mean IPDE patients						
U2 - Toxicity prior Beta(0.15, 0.85)	0.4	2.8	0.8	0.3	0	
U3 - Toxicity prior Beta(0.15, 0.85)	0.4	2.8	0.8	0.3	0	
U2 - Toxicity prior Beta(0.3, 0.7)	0.4	2.6	0.8	0.4	0	
U3 - Toxicity prior Beta(0.3, 0.7)	0.4	2.6	0.8	0.4	0	
U2 - Toxicity prior Beta(0.5, 0.5)	0.3	2.5	0.8	0.4	0	
U3 - Toxicity prior Beta(0.5, 0.5)	0.4	2.5	0.8	0.4	0	
U2 - Toxicity prior Beta(1, 1)	0.3	2.3	0.7	0.3	0	
U3 - Toxicity prior Beta(1, 1)	0.3	2.2	0.7	0.4	0	

Phase I/II Operating Characteristics

Random model - Toxicity prior comparison - allocation = two_stage - IPDE alpha = 0.3 - both utilities - N = 30

	Dose 1	Dose 2	Dose 3	Dose 4	Dose 5	Stop %
Scenario 29						
DLT rate	0.19	0.3	0.56	0.74	0.86	
Efficacy rate	0.4	0.4	0.4	0.4	0.4	
Utility 2	0.2	0.1	-0.2	-0.3	-0.5	
Utility 3	56.4	52	41.6	34.4	29.6	
OBD selection (%)						
U2 - Toxicity prior Beta(0.15, 0.85)	70.8	22.5	3.8	0.1	0	2.8
U3 - Toxicity prior Beta(0.15, 0.85)	72.8	21.6	2.7	0.2	0	2.7
U2 - Toxicity prior Beta(0.3, 0.7)	69.8	23.1	4.3	0.3	0	2.5
U3 - Toxicity prior Beta(0.3, 0.7)	72.6	21.3	3.3	0.2	0	2.6
U2 - Toxicity prior Beta(0.5, 0.5)	69.2	23.5	4.3	0.3	0	2.7
U3 - Toxicity prior Beta(0.5, 0.5)	72.2	22	3.2	0.1	0	2.5
U2 - Toxicity prior Beta(1, 1)	68.8	23.8	4.5	0.4	0	2.5
U3 - Toxicity prior Beta(1, 1)	71.8	21.8	3.5	0.4	0	2.5
Mean patients treated						
U2 - Toxicity prior Beta(0.15, 0.85)	17.9	7.5	3.4	0.6	0.1	
U3 - Toxicity prior Beta(0.15, 0.85)	17.7	7.6	3.5	0.6	0.1	
U2 - Toxicity prior Beta(0.3, 0.7)	17.3	7.4	3.9	0.9	0.1	
U3 - Toxicity prior Beta(0.3, 0.7)	17.4	7.3	3.9	0.9	0.1	
U2 - Toxicity prior Beta(0.5, 0.5)	17	7.5	4	1	0.1	
U3 - Toxicity prior Beta(0.5, 0.5)	16.9	7.6	4	0.9	0.1	
U2 - Toxicity prior Beta(1, 1)	16.9	7.6	3.9	1	0.1	
U3 - Toxicity prior Beta(1, 1)	16.9	7.6	3.9	1	0.1	
Mean IPDE patients						
U2 - Toxicity prior Beta(0.15, 0.85)	0.7	2	0.7	0.2	0	
U3 - Toxicity prior Beta(0.15, 0.85)	0.7	2	0.7	0.2	0	
U2 - Toxicity prior Beta(0.3, 0.7)	0.8	1.9	0.8	0.2	0	
U3 - Toxicity prior Beta(0.3, 0.7)	0.8	1.9	0.7	0.2	0	
U2 - Toxicity prior Beta(0.5, 0.5)	0.6	1.9	0.7	0.2	0	
U3 - Toxicity prior Beta(0.5, 0.5)	0.6	1.9	0.7	0.2	0	
U2 - Toxicity prior Beta(1, 1)	0.6	1.7	0.6	0.2	0	
U3 - Toxicity prior Beta(1, 1)	0.6	1.7	0.6	0.2	0	
Scenario 30						
DLT rate	0.05	0.08	0.17	0.27	0.54	
Efficacy rate	0.12	0.15	0.2	0.4	0.5	
Utility 2	0.1	0.1	0	0.1	0	
Utility 3	45.2	45.8	45.2	53.2	48.4	
OBD selection (%)						
U2 - Toxicity prior Beta(0.15, 0.85)	22.1	15.8	14.1	35.3	11.9	0.8
U3 - Toxicity prior Beta(0.15, 0.85)	28.2	18.8	12.1	29.9	10.2	0.8
U2 - Toxicity prior Beta(0.3, 0.7)	20.6	12.9	14.2	39.2	12.3	0.8
U3 - Toxicity prior Beta(0.3, 0.7)	25.8	16.7	12	34.2	10.5	0.8
U2 - Toxicity prior Beta(0.5, 0.5)	19.7	13	14	38.8	13.7	0.8
U3 - Toxicity prior Beta(0.5, 0.5)	25.4	17.7	11.7	33.1	11.3	0.8
U2 - Toxicity prior Beta(1, 1)	20.1	12.7	13.4	37.7	15.3	0.8
U3 - Toxicity prior Beta(1, 1)	27.3	16	12.6	32	11.3	0.8
Mean patients treated						
U2 - Toxicity prior Beta(0.15, 0.85)	5.2	4.5	5.7	8.1	6.4	
U3 - Toxicity prior Beta(0.15, 0.85)	5.1	4.6	5.8	8.1	6.4	
U2 - Toxicity prior Beta(0.3, 0.7)	4.9	4.3	5.5	8.4	6.8	
U3 - Toxicity prior Beta(0.3, 0.7)	4.8	4.3	5.5	8.5	6.8	
U2 - Toxicity prior Beta(0.5, 0.5)	4.8	4.2	5.4	8.5	7	
U3 - Toxicity prior Beta(0.5, 0.5)	4.9	4.2	5.4	8.5	7.1	
U2 - Toxicity prior Beta(1, 1)	4.9	4.1	5.2	8.3	7.4	
U3 - Toxicity prior Beta(1, 1)	4.8	4.1	5.2	8.3	7.5	
Mean IPDE patients						
U2 - Toxicity prior Beta(0.15, 0.85)	0.4	2.7	1.5	2.3	1	
U3 - Toxicity prior Beta(0.15, 0.85)	0.4	2.8	1.6	2.3	0.9	
U2 - Toxicity prior Beta(0.3, 0.7)	0.3	2.7	1.5	2.3	1	
U3 - Toxicity prior Beta(0.3, 0.7)	0.3	2.6	1.5	2.3	0.9	
U2 - Toxicity prior Beta(0.5, 0.5)	0.3	2.6	1.5	2.2	0.9	
U3 - Toxicity prior Beta(0.5, 0.5)	0.4	2.6	1.5	2.2	0.9	
U2 - Toxicity prior Beta(1, 1)	0.4	2.6	1.4	2.2	0.9	
U3 - Toxicity prior Beta(1, 1)	0.3	2.6	1.4	2.2	0.9	

Phase I/II Operating Characteristics

Random model - Toxicity prior comparison - allocation = two_stage - IPDE alpha = 0.3 - both utilities - N = 30

	Dose 1	Dose 2	Dose 3	Dose 4	Dose 5	Stop %
Scenario 31						
DLT rate	0.04	0.06	0.1	0.2	0.6	
Efficacy rate	0.2	0.5	0.5	0.5	0.5	
Utility 2	0.2	0.4	0.4	0.3	-0.1	
Utility 3	50.4	67.6	66	62	46	
OBD selection (%)						
U2 - Toxicity prior Beta(0.15, 0.85)	7.4	31	28.4	25.8	7.3	0.1
U3 - Toxicity prior Beta(0.15, 0.85)	9.7	37.4	26.8	21.5	4.5	0.1
U2 - Toxicity prior Beta(0.3, 0.7)	7.3	30.6	28.4	26	7.6	0.1
U3 - Toxicity prior Beta(0.3, 0.7)	9.7	37.2	26.6	21.8	4.6	0.1
U2 - Toxicity prior Beta(0.5, 0.5)	7.2	30.4	29.1	26.4	6.8	0.1
U3 - Toxicity prior Beta(0.5, 0.5)	9.8	38.7	26.1	21	4.3	0.1
U2 - Toxicity prior Beta(1, 1)	7.5	29.6	28.6	26.9	7.3	0.1
U3 - Toxicity prior Beta(1, 1)	9.2	38.4	27.4	20.9	4	0.1
Mean patients treated						
U2 - Toxicity prior Beta(0.15, 0.85)	4.1	5.5	6.3	8	6.1	
U3 - Toxicity prior Beta(0.15, 0.85)	4.1	5.5	6.4	7.9	6.1	
U2 - Toxicity prior Beta(0.3, 0.7)	4.1	5.4	6.4	7.9	6.3	
U3 - Toxicity prior Beta(0.3, 0.7)	4.1	5.3	6.3	8	6.3	
U2 - Toxicity prior Beta(0.5, 0.5)	4.1	5.4	6.3	7.8	6.4	
U3 - Toxicity prior Beta(0.5, 0.5)	4.1	5.3	6.3	7.9	6.5	
U2 - Toxicity prior Beta(1, 1)	4.1	5.2	6.3	7.9	6.6	
U3 - Toxicity prior Beta(1, 1)	4.1	5.2	6.2	7.8	6.6	
Mean IPDE patients						
U2 - Toxicity prior Beta(0.15, 0.85)	0.2	2.7	1.4	1.8	1.1	
U3 - Toxicity prior Beta(0.15, 0.85)	0.1	2.7	1.4	1.8	1.2	
U2 - Toxicity prior Beta(0.3, 0.7)	0.2	2.7	1.4	1.8	1.1	
U3 - Toxicity prior Beta(0.3, 0.7)	0.2	2.7	1.4	1.8	1.1	
U2 - Toxicity prior Beta(0.5, 0.5)	0.2	2.7	1.3	1.8	1.1	
U3 - Toxicity prior Beta(0.5, 0.5)	0.2	2.7	1.4	1.8	1.1	
U2 - Toxicity prior Beta(1, 1)	0.2	2.6	1.4	1.8	1.1	
U3 - Toxicity prior Beta(1, 1)	0.2	2.6	1.3	1.8	1.1	
Scenario 32						
DLT rate	0.02	0.05	0.11	0.3	0.36	
Efficacy rate	0.08	0.15	0.3	0.3	0.3	
Utility 2	0.1	0.1	0.2	0	-0.1	
Utility 3	44	47	53.6	46	43.6	
OBD selection (%)						
U2 - Toxicity prior Beta(0.15, 0.85)	10.1	14.3	34.9	25.9	14.2	0.6
U3 - Toxicity prior Beta(0.15, 0.85)	17.9	18.6	31.4	23.3	8.2	0.6
U2 - Toxicity prior Beta(0.3, 0.7)	11.7	13.7	33.8	25.7	14.5	0.6
U3 - Toxicity prior Beta(0.3, 0.7)	16.4	19	33	22.6	8.4	0.6
U2 - Toxicity prior Beta(0.5, 0.5)	10.4	14.5	35.6	24.3	14.6	0.6
U3 - Toxicity prior Beta(0.5, 0.5)	15.9	20.4	32.7	22.1	8.3	0.6
U2 - Toxicity prior Beta(1, 1)	9.9	14.6	33.9	25.1	15.9	0.6
U3 - Toxicity prior Beta(1, 1)	15.5	19	32.9	22.2	9.8	0.6
Mean patients treated						
U2 - Toxicity prior Beta(0.15, 0.85)	4	4.3	7.5	7	7.2	
U3 - Toxicity prior Beta(0.15, 0.85)	4.1	4.2	7.4	7.1	7.2	
U2 - Toxicity prior Beta(0.3, 0.7)	4	4.1	7.3	7.1	7.3	
U3 - Toxicity prior Beta(0.3, 0.7)	4	4.1	7.3	7.2	7.3	
U2 - Toxicity prior Beta(0.5, 0.5)	4	4.1	7.2	7.2	7.5	
U3 - Toxicity prior Beta(0.5, 0.5)	4	4.1	7.3	7.1	7.5	
U2 - Toxicity prior Beta(1, 1)	4	4	7	7.2	7.7	
U3 - Toxicity prior Beta(1, 1)	4	4	7.1	7.1	7.6	
Mean IPDE patients						
U2 - Toxicity prior Beta(0.15, 0.85)	0.3	2.8	1.9	2.6	1.2	
U3 - Toxicity prior Beta(0.15, 0.85)	0.3	2.8	1.9	2.6	1.1	
U2 - Toxicity prior Beta(0.3, 0.7)	0.3	2.8	1.9	2.6	1.2	
U3 - Toxicity prior Beta(0.3, 0.7)	0.3	2.8	1.9	2.6	1.2	
U2 - Toxicity prior Beta(0.5, 0.5)	0.4	2.8	1.9	2.5	1.2	
U3 - Toxicity prior Beta(0.5, 0.5)	0.3	2.8	1.9	2.6	1.2	
U2 - Toxicity prior Beta(1, 1)	0.3	2.8	1.9	2.5	1.2	
U3 - Toxicity prior Beta(1, 1)	0.3	2.8	1.9	2.5	1.2	

Phase I/II Operating Characteristics

Random model - Toxicity prior comparison - allocation = two_stage - IPDE alpha = 0.3 - both utilities - N = 30

	Dose 1	Dose 2	Dose 3	Dose 4	Dose 5	Stop %
Scenario 33						
DLT rate	0.01	0.02	0.09	0.32	0.38	
Efficacy rate	0.05	0.08	0.2	0.4	0.5	
Utility 2	0	0.1	0.1	0.1	0.1	
Utility 3	42.6	44	48.4	51.2	54.8	
OBD selection (%)						
U2 - Toxicity prior Beta(0.15, 0.85)	8.5	5.8	13.8	38.8	33.1	0
U3 - Toxicity prior Beta(0.15, 0.85)	14.5	7.5	11.7	39.1	27.2	0
U2 - Toxicity prior Beta(0.3, 0.7)	9.4	4.3	13.1	39.3	33.9	0
U3 - Toxicity prior Beta(0.3, 0.7)	13.9	6.2	12.6	38.5	28.8	0
U2 - Toxicity prior Beta(0.5, 0.5)	8.5	4.2	12.5	40	34.8	0
U3 - Toxicity prior Beta(0.5, 0.5)	12	5.6	14	39.3	29.1	0
U2 - Toxicity prior Beta(1, 1)	8	3.9	12.6	38.7	36.8	0
U3 - Toxicity prior Beta(1, 1)	12.3	6.7	13.1	36.7	31.2	0
Mean patients treated						
U2 - Toxicity prior Beta(0.15, 0.85)	3.5	3.6	5	8	9.9	
U3 - Toxicity prior Beta(0.15, 0.85)	3.5	3.5	5	8.2	9.7	
U2 - Toxicity prior Beta(0.3, 0.7)	3.5	3.4	4.9	8.2	10	
U3 - Toxicity prior Beta(0.3, 0.7)	3.5	3.4	4.9	8.1	10.1	
U2 - Toxicity prior Beta(0.5, 0.5)	3.5	3.4	4.8	8.1	10.2	
U3 - Toxicity prior Beta(0.5, 0.5)	3.5	3.4	4.8	8	10.2	
U2 - Toxicity prior Beta(1, 1)	3.5	3.4	4.7	7.9	10.5	
U3 - Toxicity prior Beta(1, 1)	3.5	3.4	4.7	7.9	10.4	
Mean IPDE patients						
U2 - Toxicity prior Beta(0.15, 0.85)	0.2	2.6	1.4	2.7	1	
U3 - Toxicity prior Beta(0.15, 0.85)	0.2	2.6	1.4	2.8	1	
U2 - Toxicity prior Beta(0.3, 0.7)	0.2	2.6	1.4	2.8	1	
U3 - Toxicity prior Beta(0.3, 0.7)	0.2	2.6	1.4	2.7	1	
U2 - Toxicity prior Beta(0.5, 0.5)	0.2	2.6	1.4	2.7	1	
U3 - Toxicity prior Beta(0.5, 0.5)	0.2	2.6	1.4	2.7	1	
U2 - Toxicity prior Beta(1, 1)	0.2	2.6	1.3	2.6	1	
U3 - Toxicity prior Beta(1, 1)	0.2	2.6	1.4	2.6	0.9	
Scenario 34						
DLT rate	0.05	0.13	0.3	0.36	0.42	
Efficacy rate	0.05	0.1	0.3	0.32	0.35	
Utility 2	0	0	0	0	-0.1	
Utility 3	41	40.8	46	44.8	44.2	
OBD selection (%)						
U2 - Toxicity prior Beta(0.15, 0.85)	16.7	13.8	44.5	17.2	6	1.8
U3 - Toxicity prior Beta(0.15, 0.85)	23.8	15.8	39.2	15.6	3.6	2
U2 - Toxicity prior Beta(0.3, 0.7)	17.9	12.9	42.4	18	7	1.8
U3 - Toxicity prior Beta(0.3, 0.7)	23.6	15.6	37.9	16.3	4.8	1.8
U2 - Toxicity prior Beta(0.5, 0.5)	14.8	13.2	42.2	20.1	7.8	1.9
U3 - Toxicity prior Beta(0.5, 0.5)	22.9	15	37.4	18	4.9	1.8
U2 - Toxicity prior Beta(1, 1)	16.1	12.6	40.8	20.8	7.8	1.9
U3 - Toxicity prior Beta(1, 1)	22.9	14.3	36.8	19	5.1	1.9
Mean patients treated						
U2 - Toxicity prior Beta(0.15, 0.85)	5.4	5.7	9.3	5.6	3.7	
U3 - Toxicity prior Beta(0.15, 0.85)	5.3	5.7	9.5	5.6	3.7	
U2 - Toxicity prior Beta(0.3, 0.7)	5.1	5.3	9	6.2	4.2	
U3 - Toxicity prior Beta(0.3, 0.7)	5	5.4	9	6.2	4.2	
U2 - Toxicity prior Beta(0.5, 0.5)	4.9	5.1	8.8	6.5	4.4	
U3 - Toxicity prior Beta(0.5, 0.5)	5	5.1	8.8	6.5	4.4	
U2 - Toxicity prior Beta(1, 1)	4.9	4.9	8.7	6.6	4.7	
U3 - Toxicity prior Beta(1, 1)	4.9	4.8	8.7	6.5	4.8	
Mean IPDE patients						
U2 - Toxicity prior Beta(0.15, 0.85)	0.6	3.2	1.9	1.7	0.6	
U3 - Toxicity prior Beta(0.15, 0.85)	0.6	3.2	1.9	1.7	0.6	
U2 - Toxicity prior Beta(0.3, 0.7)	0.5	3	1.9	1.8	0.7	
U3 - Toxicity prior Beta(0.3, 0.7)	0.5	3	1.9	1.8	0.7	
U2 - Toxicity prior Beta(0.5, 0.5)	0.4	3	1.8	1.7	0.7	
U3 - Toxicity prior Beta(0.5, 0.5)	0.4	3	1.9	1.7	0.7	
U2 - Toxicity prior Beta(1, 1)	0.4	2.9	1.8	1.7	0.6	
U3 - Toxicity prior Beta(1, 1)	0.4	2.9	1.8	1.7	0.6	

Phase I/II Operating Characteristics

Random model - Toxicity prior comparison - allocation = two_stage - IPDE alpha = 0.3 - both utilities - N = 30

	Dose 1	Dose 2	Dose 3	Dose 4	Dose 5	Stop %
Scenario 35						
DLT rate	0.03	0.09	0.3	0.4	0.45	
Efficacy rate	0.2	0.45	0.45	0.45	0.45	
Utility 2	0.2	0.4	0.1	0.1	0	
Utility 3	50.8	63.4	55	51	49	
OBD selection (%)						
U2 - Toxicity prior Beta(0.15, 0.85)	11.9	36.4	36.6	11.4	3.6	0.1
U3 - Toxicity prior Beta(0.15, 0.85)	15.8	42	30.2	9.7	2.2	0.1
U2 - Toxicity prior Beta(0.3, 0.7)	12.8	37.8	32.1	13.1	4.1	0.1
U3 - Toxicity prior Beta(0.3, 0.7)	15	43	28.7	10.8	2.4	0.1
U2 - Toxicity prior Beta(0.5, 0.5)	12.2	36.5	32.7	14.1	4.4	0.1
U3 - Toxicity prior Beta(0.5, 0.5)	16.1	42.1	27.9	11	2.8	0.1
U2 - Toxicity prior Beta(1, 1)	11.8	37.7	31.9	14.1	4.4	0.1
U3 - Toxicity prior Beta(1, 1)	15.7	42.4	27.9	10.8	3.1	0.1
Mean patients treated						
U2 - Toxicity prior Beta(0.15, 0.85)	4.6	8	9	5.3	3.2	
U3 - Toxicity prior Beta(0.15, 0.85)	4.6	7.9	9	5.2	3.2	
U2 - Toxicity prior Beta(0.3, 0.7)	4.4	7.7	8.6	5.8	3.5	
U3 - Toxicity prior Beta(0.3, 0.7)	4.5	7.7	8.6	5.7	3.4	
U2 - Toxicity prior Beta(0.5, 0.5)	4.4	7.6	8.4	6	3.6	
U3 - Toxicity prior Beta(0.5, 0.5)	4.4	7.6	8.4	5.9	3.7	
U2 - Toxicity prior Beta(1, 1)	4.3	7.5	8.3	6.1	3.8	
U3 - Toxicity prior Beta(1, 1)	4.4	7.5	8.2	6.1	3.8	
Mean IPDE patients						
U2 - Toxicity prior Beta(0.15, 0.85)	0.3	3	1.6	1.4	0.5	
U3 - Toxicity prior Beta(0.15, 0.85)	0.3	3	1.6	1.4	0.5	
U2 - Toxicity prior Beta(0.3, 0.7)	0.3	2.9	1.6	1.5	0.5	
U3 - Toxicity prior Beta(0.3, 0.7)	0.3	3	1.6	1.5	0.5	
U2 - Toxicity prior Beta(0.5, 0.5)	0.3	3	1.5	1.4	0.5	
U3 - Toxicity prior Beta(0.5, 0.5)	0.3	2.9	1.5	1.5	0.6	
U2 - Toxicity prior Beta(1, 1)	0.3	3	1.5	1.4	0.5	
U3 - Toxicity prior Beta(1, 1)	0.3	3	1.5	1.4	0.5	
Scenario 36						
DLT rate	0.13	0.29	0.35	0.4	0.43	
Efficacy rate	0.05	0.2	0.22	0.3	0.35	
Utility 2	-0.1	-0.1	-0.1	-0.1	-0.1	
Utility 3	37.8	40.4	39.2	42	43.8	
OBD selection (%)						
U2 - Toxicity prior Beta(0.15, 0.85)	23.4	38.1	14.6	6.6	2.6	14.7
U3 - Toxicity prior Beta(0.15, 0.85)	29.7	36.7	11.2	6	1.4	15
U2 - Toxicity prior Beta(0.3, 0.7)	20.6	36.9	15.1	9.7	3.4	14.3
U3 - Toxicity prior Beta(0.3, 0.7)	27.4	36.8	12.2	7.2	2.2	14.2
U2 - Toxicity prior Beta(0.5, 0.5)	20.5	35.3	17.1	8.5	4.3	14.3
U3 - Toxicity prior Beta(0.5, 0.5)	26.2	36.6	13.4	6.8	3	14
U2 - Toxicity prior Beta(1, 1)	20.4	34.2	16	10.1	4.9	14.4
U3 - Toxicity prior Beta(1, 1)	25.9	34.3	13.4	8.5	3.4	14.5
Mean patients treated						
U2 - Toxicity prior Beta(0.15, 0.85)	9.1	8.4	5.7	3.1	1.9	
U3 - Toxicity prior Beta(0.15, 0.85)	9.2	8.3	5.7	3.2	1.9	
U2 - Toxicity prior Beta(0.3, 0.7)	8.6	7.8	6	3.7	2.2	
U3 - Toxicity prior Beta(0.3, 0.7)	8.6	7.8	6	3.7	2.2	
U2 - Toxicity prior Beta(0.5, 0.5)	8.5	7.3	6	4.1	2.4	
U3 - Toxicity prior Beta(0.5, 0.5)	8.5	7.3	6	4.1	2.5	
U2 - Toxicity prior Beta(1, 1)	8.3	7.3	5.9	4.2	2.7	
U3 - Toxicity prior Beta(1, 1)	8.3	7.3	5.9	4.2	2.7	
Mean IPDE patients						
U2 - Toxicity prior Beta(0.15, 0.85)	0.5	3.5	1.1	1	0.3	
U3 - Toxicity prior Beta(0.15, 0.85)	0.5	3.4	1.1	1	0.3	
U2 - Toxicity prior Beta(0.3, 0.7)	0.5	3.3	1.2	1.2	0.3	
U3 - Toxicity prior Beta(0.3, 0.7)	0.5	3.2	1.2	1.1	0.3	
U2 - Toxicity prior Beta(0.5, 0.5)	0.4	3.1	1.2	1.1	0.3	
U3 - Toxicity prior Beta(0.5, 0.5)	0.4	3.1	1.2	1.1	0.3	
U2 - Toxicity prior Beta(1, 1)	0.4	2.9	1.1	1	0.3	
U3 - Toxicity prior Beta(1, 1)	0.4	2.9	1	1	0.3	

Phase I/II Operating Characteristics

Random model - Toxicity prior comparison - allocation = two_stage - IPDE alpha = 0.3 - both utilities - N = 30

	Dose 1	Dose 2	Dose 3	Dose 4	Dose 5	Stop %
Scenario 37						
DLT rate	0.08	0.3	0.4	0.45	0.5	
Efficacy rate	0.05	0.2	0.2	0.2	0.2	
Utility 2	0	-0.1	-0.2	-0.2	-0.3	
Utility 3	39.8	40	36	34	32	
OBD selection (%)						
U2 - Toxicity prior Beta(0.15, 0.85)	29.5	47.1	12.9	2.6	0.9	7
U3 - Toxicity prior Beta(0.15, 0.85)	36	44.4	9.8	2.1	0.7	7
U2 - Toxicity prior Beta(0.3, 0.7)	28.9	44.5	14.2	4.7	1.1	6.6
U3 - Toxicity prior Beta(0.3, 0.7)	35.1	43.8	10.2	3.6	0.9	6.4
U2 - Toxicity prior Beta(0.5, 0.5)	28.9	43.6	15.7	3.8	1.7	6.3
U3 - Toxicity prior Beta(0.5, 0.5)	35.2	43.3	10.9	3	1.1	6.5
U2 - Toxicity prior Beta(1, 1)	26.6	45.2	16.2	4.2	1.6	6.2
U3 - Toxicity prior Beta(1, 1)	34.3	43.9	11.9	2.7	1.1	6.1
Mean patients treated						
U2 - Toxicity prior Beta(0.15, 0.85)	8.4	10.1	6.5	2.7	1.3	
U3 - Toxicity prior Beta(0.15, 0.85)	8.5	10.2	6.4	2.6	1.3	
U2 - Toxicity prior Beta(0.3, 0.7)	8.1	9.8	6.5	3.3	1.5	
U3 - Toxicity prior Beta(0.3, 0.7)	8.1	9.7	6.5	3.2	1.5	
U2 - Toxicity prior Beta(0.5, 0.5)	7.8	9.6	6.5	3.6	1.6	
U3 - Toxicity prior Beta(0.5, 0.5)	7.8	9.5	6.6	3.7	1.6	
U2 - Toxicity prior Beta(1, 1)	7.7	9.4	6.5	3.8	1.8	
U3 - Toxicity prior Beta(1, 1)	7.7	9.3	6.5	3.8	1.9	
Mean IPDE patients						
U2 - Toxicity prior Beta(0.15, 0.85)	0.8	4	1.3	1	0.2	
U3 - Toxicity prior Beta(0.15, 0.85)	0.8	4	1.3	1	0.2	
U2 - Toxicity prior Beta(0.3, 0.7)	0.8	3.7	1.3	1	0.3	
U3 - Toxicity prior Beta(0.3, 0.7)	0.8	3.8	1.3	1	0.3	
U2 - Toxicity prior Beta(0.5, 0.5)	0.7	3.6	1.3	1	0.3	
U3 - Toxicity prior Beta(0.5, 0.5)	0.6	3.6	1.3	1	0.2	
U2 - Toxicity prior Beta(1, 1)	0.6	3.4	1.2	1	0.2	
U3 - Toxicity prior Beta(1, 1)	0.6	3.4	1.2	1	0.2	